

Forensic Medicine

World of Revision

Marrow

MARROW

Instructions

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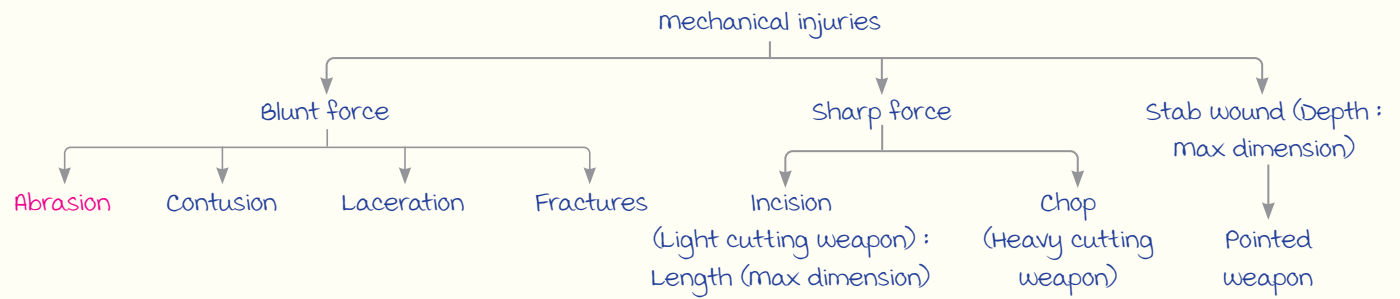
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TRAUMATOLOGY

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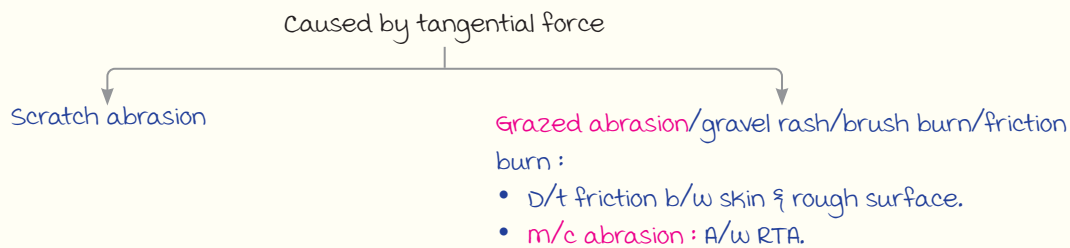


Abrasion

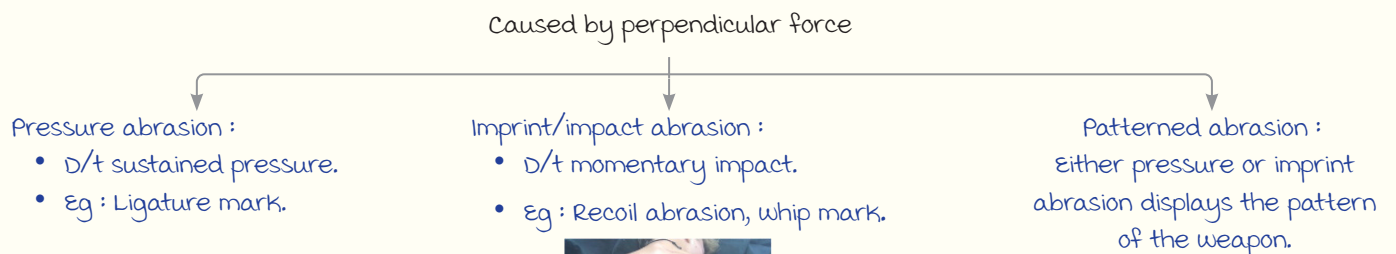
00:01:14

- medicolegally significant.
- Superficial injury → No bleeding/scarring.
- Heals in : 1 week.

Types :



multiple scratches
over a wide area



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Epithelial Tag :

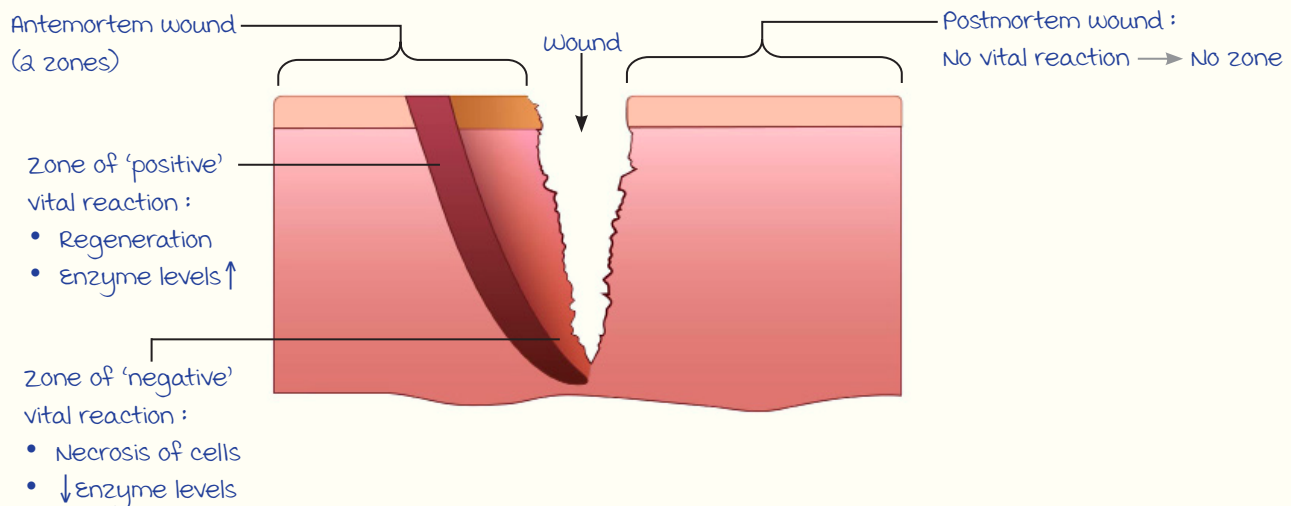
- Epithelium is scraped off & heaped.
- Indicates tail end of the abrasion.
- Determines **direction** of force.

Aging of Abrasion :

Colour of swab → Gives **time since injury** (TSI).

Mnemonic **R³B³**.

- **Raw** : < 12 hours.
- **Reddish** : > 12 hours.
- **Reddish brown** : 2 - 3 days.
- **Brown** : 4 - 5 days.
- **Black** : 6 - 7 days.

Antemortem vs. Postmortem Wound :Contusion/Bruise

00:05:44

Blunt force trauma

↓

Rupture of dermal vessels

↓

Extravasation of blood.



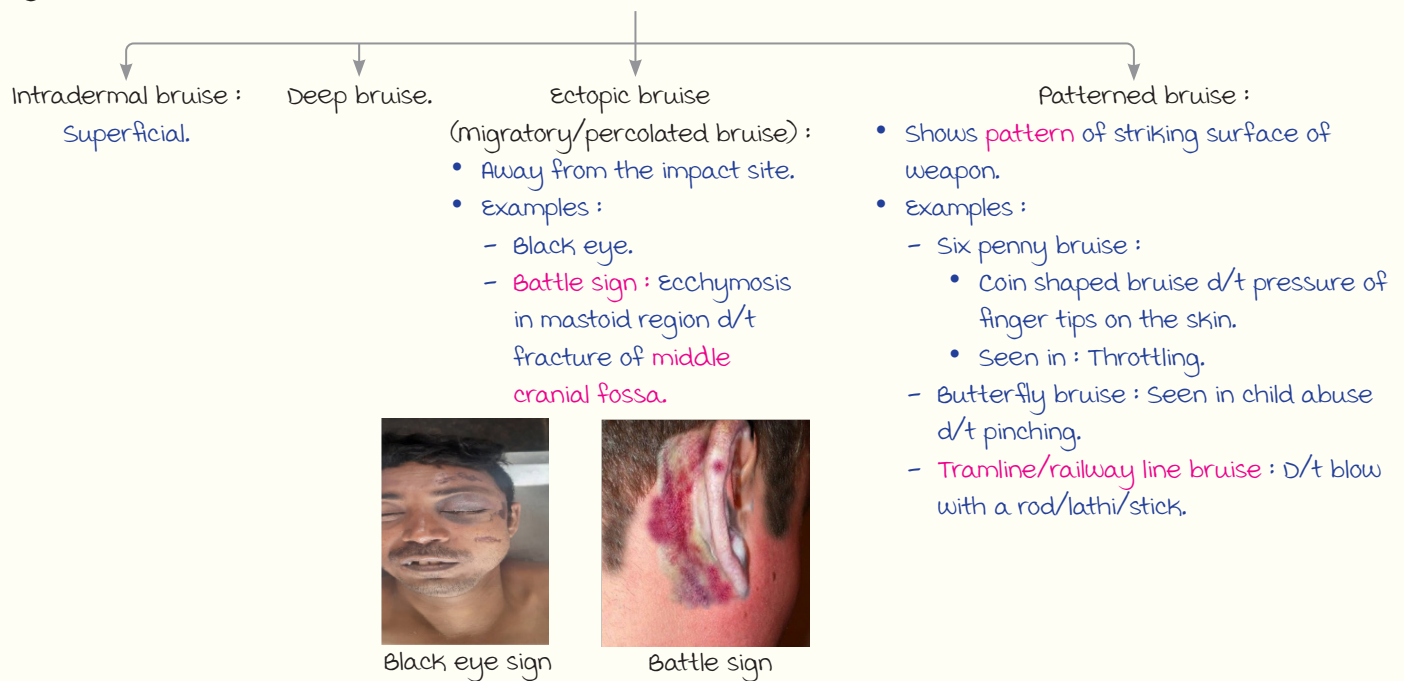
Contusion

Contusion vs Hypostasis :

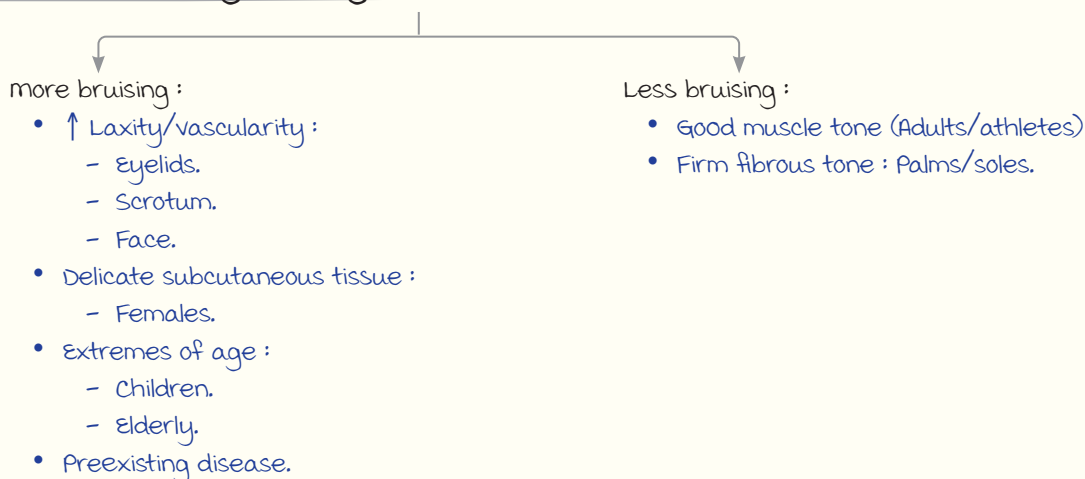
	Contusion	Hypostasis
Blanching	⊖	⊕
Extravasation	⊕	⊖
Colour changes	⊕	⊖
margins	Ill-defined	Well-defined
Sites	Anywhere on body	Dependent areas

Types :

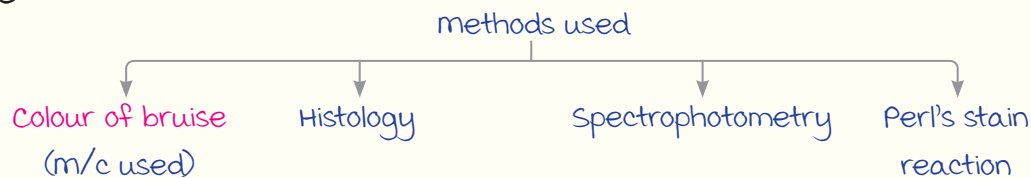
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True Bruise vs Artificial Bruise 💀 :

	True bruise	Artificial bruise
Cause	Trauma	Irritant plant extract (Plumbago, semicarpus, calotropis)
Site	Anywhere on body	Accessible parts of body
Vesication /blisters	Absent	Present (D/t inflammatory reaction)
Content	Blood	Inflammatory fluid → Acid serum
Itching	Absent	Present

Factors Affecting Bruising :

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Aging of Contusion 🩸 :

Colour of the bruise :

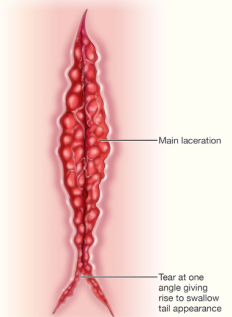
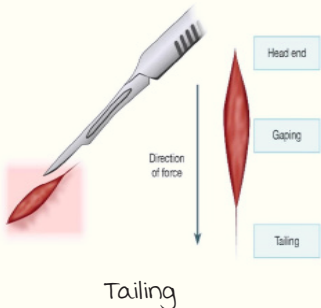
Ageing of bruise by color		
Type of hemoglobin	Color	Age of bruise
Oxyhemoglobin	Red	Fresh
Deoxyhemoglobin	Blue	Few hrs to 3 days
Hemosiderin	Brown	4 days
Biliverdin	Green	5-6 days
Bilirubin	Yellow	7-12 days

multiple bruises of different colour → Sign of child abuse (Battered baby syndrome).

Laceration, Incision & Stab Wound

00:12:43

Laceration vs. Incision :

	Laceration (Tear)	Incision (Cut)
margins	Irregular	Clean cut
Feature	Swallow tails	Tailing (Direction of force can be assessed)
Tissue bridges	⊕	⊖
Floor (Hair bulb, vessels & nerves)	Crushed	Cut
Images	  Swallow tailing	  Tailing

Types of Lacerations :

----- Active space -----

Split laceration 💀 :

- AKA **incised looking laceration**
- mechanism : Skin **crushed** b/w two hard objects i.e. **bony prominences**.



Split laceration : Skull



Avulsion laceration 💀 :

- Shearing force (Tangential) : Separation of skin from deeper tissues → **Flaying**
- E.g. : Degloving injury, scalping injury.



Degloving injury

Tear laceration :

Caused by semi-sharp objects.

Stretch laceration :

D/t overstretching of skin.

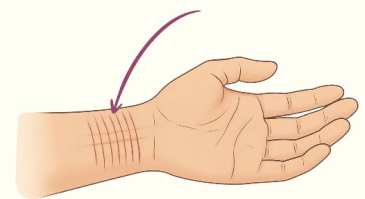
Incised Wounds :

Lacerated looking incision :

- Seen in areas with skin folds (Scrotum, axilla).
- Injury with knife having serrated edges.

Hesitation cuts :

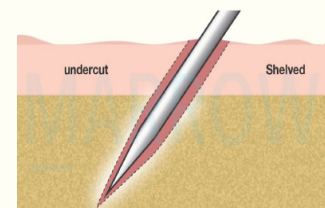
- AKA Tentative cut/intentional cut/feeler's strokes/trial cuts.
- multiple, superficial, linear cuts.
- Site : **Accessible parts of the body**.
- Indicates **suicidal attempt**.



Hesitation cuts

Bevelling :

- Blade enters obliquely into the skin → undermined edges.
- Indicates **homicide**.



Bevelling

Stab Wound 💀 :

Punctured wound :

Stab injury into tissue.

Penetrating wound :

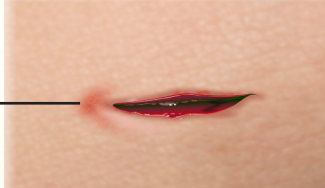
Only entry wound seen
(Into body cavity).

Perforating wound :

Entry & exit wounds
seen.

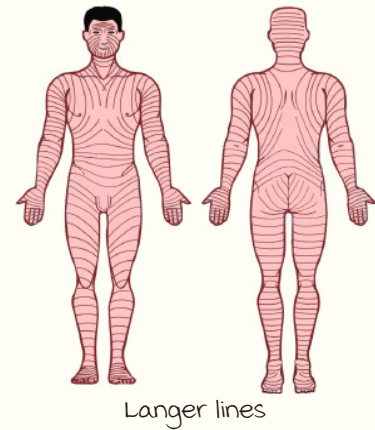
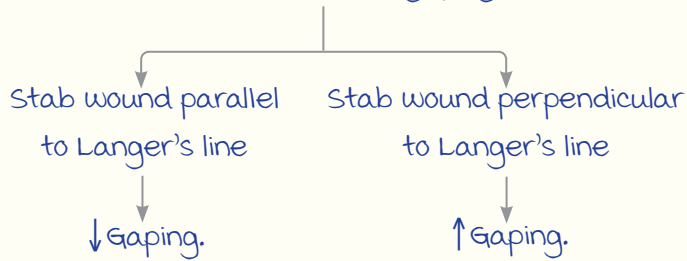
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Type of weapon based on shape of stab wound :

Shape of wound	- Wedge/triangle/pear shape - Fish tailing	Elliptical shape
Weapon	Single edge knife	Double edge knife
Image		 Hilt mark : Patterned abrasion/bruise - Seen in : Complete penetration - Helps determine : - Direction of force - Type of weapon

Lines of langer/cleavage lines :

- Represent the arrangement of collagen fibers.
- Determine the extent of gaping.



Hara-kiri/seppuku :

- Suicidal stab wound of the abdomen.
- Cause of death : Evisceration → Circulatory collapse.

Concealed punctured wounds :

Seen in :

- Axilla.
- Nostrils.
- Fontanelles.
- Nape of the neck.



Stab injury to the heart :

Fatality of stab injury : RA > LV (Fatality $\propto \frac{1}{\text{Thickness of heart chamber}}$)

Chop Wound & Defence injuries

00:24:43

----- Active space -----

Chop Wounds :

- Deep gaping wound caused by heavy sharp weapon.
- margins : Regular.
- Floor : Crushing + fracture of bone.
- usually suggestive of **homicide**.



Heavy sharp weapon



Chop wound : Cut fracture

Defence Cuts :

- Indicates **homicide**.
- Types :

Active :

Seen at palm, 1st web space.

Passive :

Seen at : Forearm, wrist.



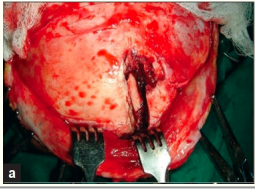
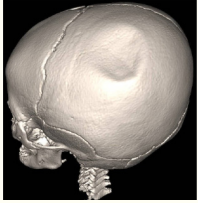
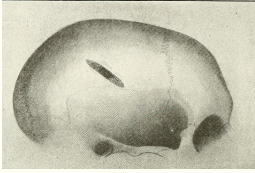
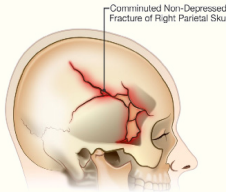
Regional Injuries

00:26:42

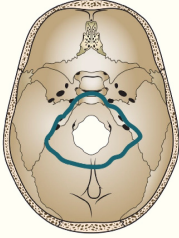
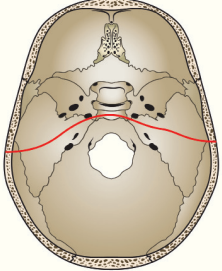
Skull vault Fractures :

Fracture	Image
Fissure fracture : • m/c type of skull fracture • Caused d/t weapon with broad striking surface	

----- Active space -----

Fracture	Image
Depressed fracture : - Caused due to weapon with smaller striking surface (Hammer) - Also known as signature fracture (Weapon can be identified)	
Pond fracture : - Variant of depressed fracture - Also known as indented fracture/ping pong fracture - Seen in infants/newborn/child (< 4 years) → Soft skull	
Gutter fracture 🦴 : Associated with oblique bullet	
Comminuted fracture : - AKA spiderweb/mosaic fracture - Multiple fractured segments without displacement	
Diastatic fracture : AKA sutural fractures as the fracture line is along the sutures of the skull	

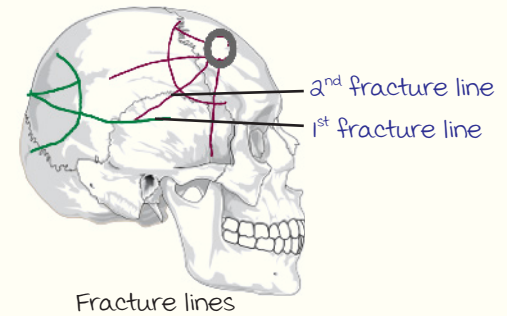
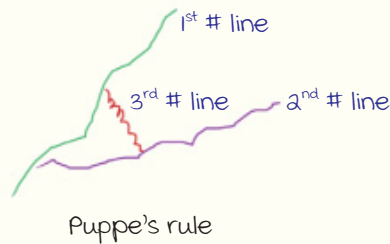
Skull Base Fractures :

Type	Weapon/mechanism	Characteristics
Ring fracture 	- Fall from height : - Lands on feet - Lands on buttock. - Heavy weight on the head	Fracture in base of skull : Around foramen magnum
Type I hinge fracture 	-	- Fracture line from temporal region reaches opposite side - AKA motorcyclist fracture (Type I hinge #)

Puppe's Rule :

----- Active space -----

- The new fracture line will never cross previous fracture line.
- Sequencing the fracture lines due to blows.

Coup & Contrecoup Injuries :

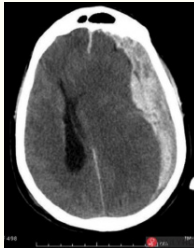
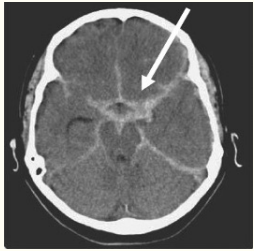
	Coup injury	Contrecoup injury
Location	Injury at site of impact	Injury opposite to site of impact
Severity	mild/nil	Severe
Occipital impact	Occipital lobe contusion	Frontal lobe contusion (m/c site of contrecoup injury)
Frontal impact	Frontal lobe contusion	Nil
Temporal impact	Temporal lobe of one side	• Temporal lobe of opposite side • Contralateral surface of ipsilateral lobe

Intracranial Hemorrhages (ICH) 🩸

00:33:40

	Extradural hemorrhage (EDH)	Subdural hemorrhage (SDH)	Subarachnoid hemorrhage (SAH)
Features	• Least common ICH • Trauma to temporo-parietal region/pterion (Type of coup injury) ↓ Fracture of temporal bone ↓ Rupture of middle meningeal artery ↓ Bleed in the extradural space ↓ Brainstem compression ↓ Death d/t respiratory failure. • min. 35 mL blood → Clinical features • Does not cross suture lines	• Rupture of bridging veins/dural venous sinuses. • Risk factors : mnemonic ABC - Aged person with minor trauma - Boxers - Child abuse (Shaken baby syndrome) • Crosses suture lines	• m/c ICH • Causes : mnemonic BATS - Berry aneurysm rupture - AV malformation rupture - Trauma - Stroke • Impending rupture → Thunderclap headache • Aneurysm rupture → Seizures/coma • Lumbar puncture → Xanthochromia

----- Active space -----

	Extradural hemorrhage (EDH)	Subdural hemorrhage (SDH)	Subarachnoid hemorrhage (SAH)
Imaging	 Biconvex opacity	 Crescentic (Concavo-convex opacity)	
Autopsy findings 💀		 Hemorrhage clears after pouring water	 Hemorrhage remains intact after pouring water

Lucid Interval :

- Period of consciousness b/w 2 periods of unconsciousness.
- Seen in **EDH** > SDH.

medicolegal importance :

- Patient can provide valid evidence, will & is criminally liable.
- Death d/t failure in diagnosing lucid interval : medical negligence.

↓
Punishable under 106(I) BNS.

Note :m/c intracranial **lesion** : Cerebral contusion.

Transportation Injuries

00:39:29

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Injury to Pedestrians :

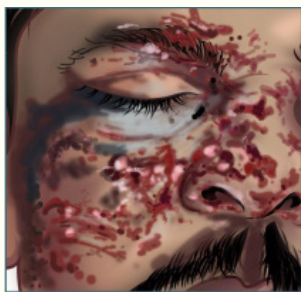


Injury to vehicle Occupants :

S. No	Impact	Injuries
1.	Windshield	Sparrow foot injuries : multiple cut lacerations to exposed body part d/t broken glass.
2.	Whiplash	Hyperextension (more dangerous) f/b hyperflexion or vice versa of neck
3.	Seat belt	• Seat belt bruise. • Organ injury : mesentery > Small intestine. • Chance fracture : Transverse vertebral fracture (d/t sudden hyperflexion).
4.	Dashboard	Dashboard fracture : Posterior dislocation of hip.
5.	Steering wheel	• Patterned bruise. • Fractures (Sternum). • Aortic injury (Ladder-rung tears).
6.	Car pedals	Ankle fracture.

Injuries sustained by :

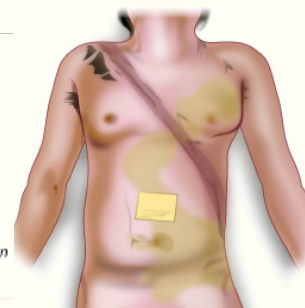
- Driver only
- Front seat passenger only
- Both



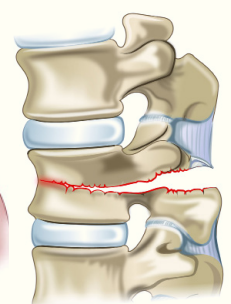
Sparrow foot mark injury



Whiplash injury



Seat belt bruise



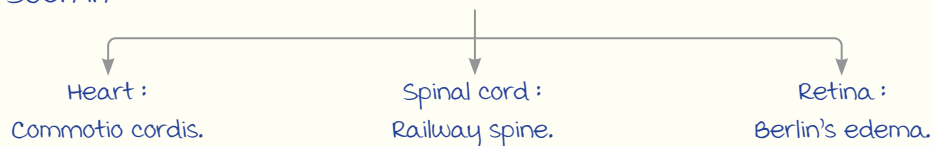
Chance fracture

----- Active space -----

Note :

• Concussion :

- Temporary stoppage of functions.
- Seen in :



• Beck's triad :

muffled heart sounds



↑ JVP

Hypotension

- Seen in cardiac tamponade.
- mx : Pericardiocentesis.

Thermal Injuries

00:44:44

	Cold injury	Heat injury
General effects	Hypothermia	Heat-related syndromes
Local effects	1. Dry cold : Frost bite 2. moist cold : Trench foot/immersion foot	1. Dry heat : Burns 2. moist heat : Scalds

Hypothermia :

- Body core temperature $< 35^{\circ}\text{C}$.
- Commonly affects elderly, newborns, alcoholics, hypothyroid patients.
- $< 32^{\circ}\text{C}$ → Shivering stops.
- $< 30^{\circ}\text{C}$ → Hypothalamus stops functioning.
- 🩸 Severe hypothermia → Paradoxical undressing.
→ Hide & die syndrome/terminal burrowing.



Trench foot

Autopsy findings :

- Pink hypostasis.
- Wischniewski's bleeding spot in stomach.

Treatment : Rewarming.

Frost Bite :

D/t exposure to freezing temperature.

Site : Periphery (Finger, feet, tip of nose/ear lobes).

Rx : Rewarming → $37^{\circ} - 40^{\circ}\text{C}$.

Frost bite

Heat Related Syndromes :

----- Active space -----

	Pathology & C/F	Thermo regulation	Core body temperature
Heat cramps	Sweating causes loss of sodium → muscle cramps.	Intact.	Normal.
Heat syncope	↑ Sweating (moist skin) → Loss of sodium and water → Cerebral hypoperfusion → Hypotension & syncope.		
Heat stroke (Fatal)	Features : 1. Core body temperature $> 40.5^{\circ}\text{C}$. 2. CNS dysfunction. 3. Clinical findings : - Skin : Dry & hot - Sweating usually absent. 4. Autopsy : Post mortem calorificity.	Impaired.	↑↑↑ ($> 40.5^{\circ}\text{C}/105^{\circ}\text{F}$)

Burns

00:49:52

min. 44°C for 5 - 6 hrs → Burns (↑ Temperature → ↓ Duration).

Burns vs Scalds :

	Burns	Scalds
Charring & singeing	⊕	⊖
Soddened & bleached skin	⊖	⊕
Ulceration	⊖	⊖
Lines of blisters	⊖	⊕
Splashing	⊖	⊕
Clothings	Burnt	Intact

Note :

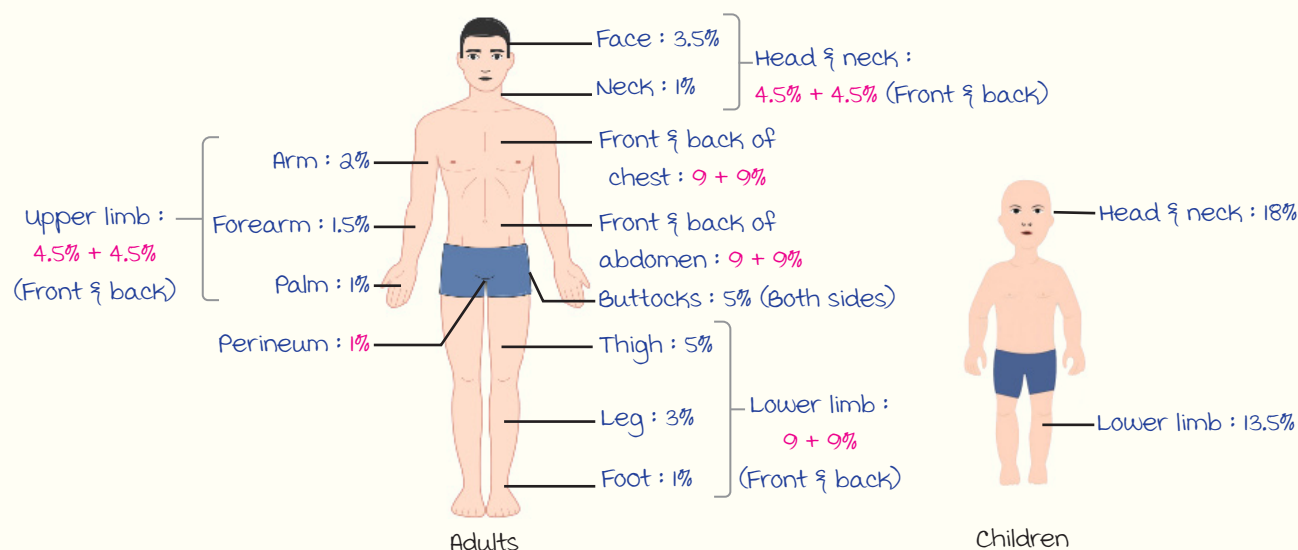
Chemical burns :

- ulceration ⊕.
- Blisters ⊖.

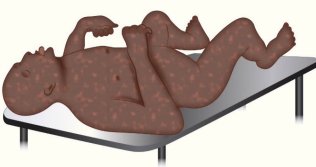
Estimation of Burnt Surface Area :

1. Rule of Wallace/Rule of 9.
2. Lund & Browders chart : Best for children.
3. Rule of palm : Size of burnt area = Size of palm = Approximately 1%.

----- Active space -----

Rule of Wallace :Autopsy Findings in Antemortem Burns :

Non-specific findings (Heat artefacts) 💀 :

	Pathogenesis	Features	Image
Heat stiffening	muscle exposed to heat $> 65^{\circ}\text{C}$ ↓ Protein coagulation ↓ Stiffening	AKA Boxer's attitude/pugilistic attitude/fencer's attitude/defence attitude	
Heat rupture	Skin exposed to heat ↓ Cracking & splitting of skin	Difference from incised wound : • Large & irregular • Intact vessels & nerves ↓ Pale	
Heat hematoma	Blood clot in the extradural space	• Chocolate brown • Honey comb appearance	
Heat fracture	Fracture of long bone/skull bones	• Long bone fracture ↓ Street & avenue fracture • Skull bone fracture ↓ Spiderweb fracture (Sides of skull)	—

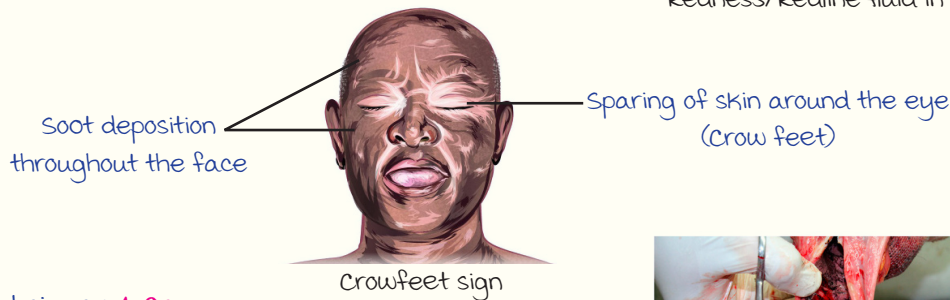
Specific signs : Features of **antemortem** burns.

External signs :

1. **Crow feet sign.**
 2. **Fluid in blisters.**
 3. **Inflammation signs** : Indicate healing.
 4. **Redness/redline.**
 5. **Enzymes** ↑ ↑.
- mnemonic : **FIRE.**



Redness/Redline fluid in blisters



Internal signs : **4 Cs**

- **Carbon** deposition (Soot) in airway.
- ↑ **Cyanide** & ↑ **CarboxyHb** (> 10 g%) in blood.
- **Curling** ulcer.



Carbon (Soot) deposition in airway

Electrical & Lightning Injuries

01:00:43

Electrocution :

Factors affecting :

- Type of current : AC more harmful than DC.
 - Voltage.
 - **Amperage** : Hold on effect (15 - 20 mA) → Tetanoid/hold-on spasms → Death.
 - Resistance :
 - Inversely proportional to magnitude of injury.
 - **Dry skin** (maximum) > bone > moist skin (**No entry wound**).
- Directly proportional to magnitude of injury.

Cause of death :

- **Cardiac arrhythmias** (m/c).
- Respiratory failure.

Electrical Injury Burns 🦴 :

Burns of low voltage current	Burns of high voltage current
• Firm contact. • Joule burn/endogenous burn : - Central depressed area. - Peripheral raised margins. • metallisation : Deposition of metallic ions in the entry wound.	• Flash burn : Diffuse burn. • Crocodile burn : multiple punctate lesions.



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Lightning Injuries :

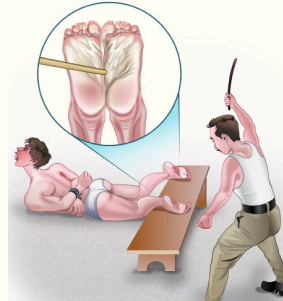
Filigree burns 💔 :

- Burns d/t lightning injury.
- Also known as :
 - Lichtenberg marking.
 - Arborescent burns.
 - Ferning.
 - Keraunographic markings.

Torture Methods 💔

01:04:23

Telefona : Repeated slapping over the ears Falanga/bastinado : Beating over the soles Saw horse : Forced straddling



Dry submarine : Plastic bag asphyxiation



Wet submarine : Forced immersion of head under water



Dunking : Forced immersion of whole body under water



El planton : Prolonged standing



Cattle's prod : Electric shock to genitalia



Hog tying : Tying wrist & ankle together in prone position



Water boarding : Water is poured over a cloth covering the face → Asphyxiation



Strappado/la bandera/palestinian hanging :
Suspension of victim by wrist



Parrot's perch : Tying upper & lower limb
with suspension of body



----- Active space -----

murcielago : Suspension by ankle.

Black slave : Inserting hot metal rod into
anus.

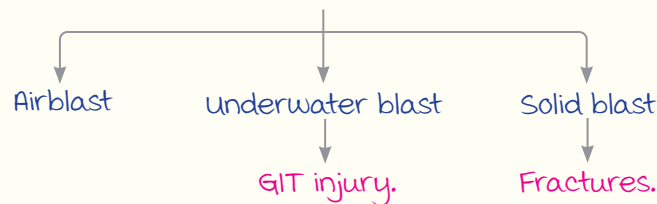
Bellary : Putting chilly powder into anus.

Belana : Rolling a heavy stick over the
thighs/legs.

Explosion Injuries

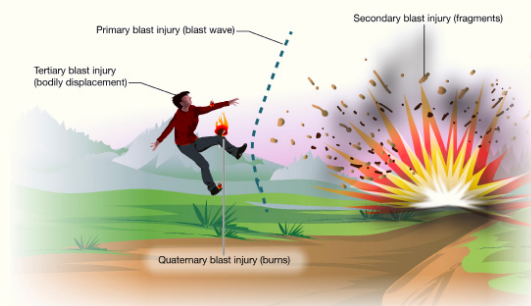
01:06:37

Types :



Airblast 💀 :

Blast Injury	Cause	Organ Injured
Primary	Blast wave	Ear (m/c), Lung (most fatal) → Blast lung
Secondary (m/c injury)	Flying missiles/projectiles	Marshal's triad : 1. Abrasion 2. Contusion 3. Laceration (Punctate) } Same surface
Tertiary	wind/victim displacement	-
Quaternary	miscellaneous	- Traumatic asphyxia - Burns



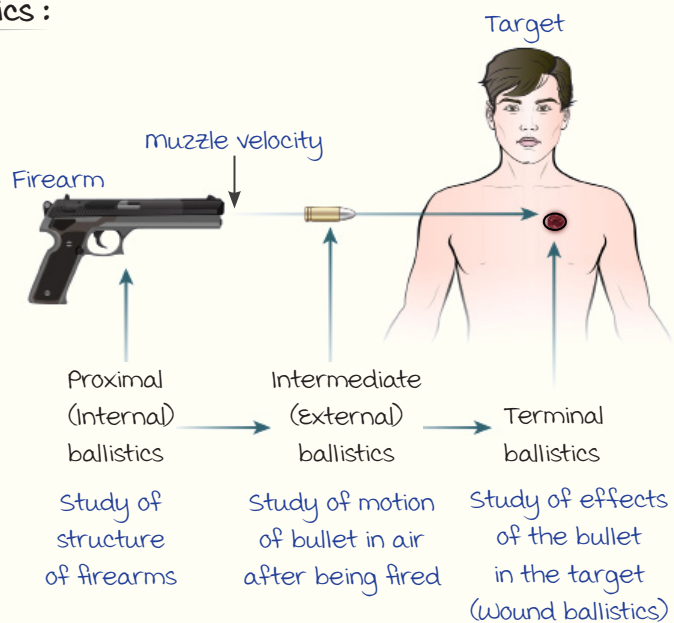
Air blast

FORENSIC BALLISTICS

Basics of Firearm

00:00:10

Types of Ballistics :

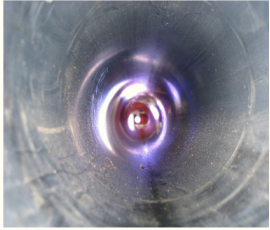


Parts of a Firearm :



Classification :

Based on the inner surface of the barrel → visualized by **helixometer**.

	Rifled gun (military rifle, pistol, revolver)	Smooth-bored gun/Shotgun
Inner surface of barrel	Rifling ⊕ (Spiral grooves in the inner surface) 	Smooth 

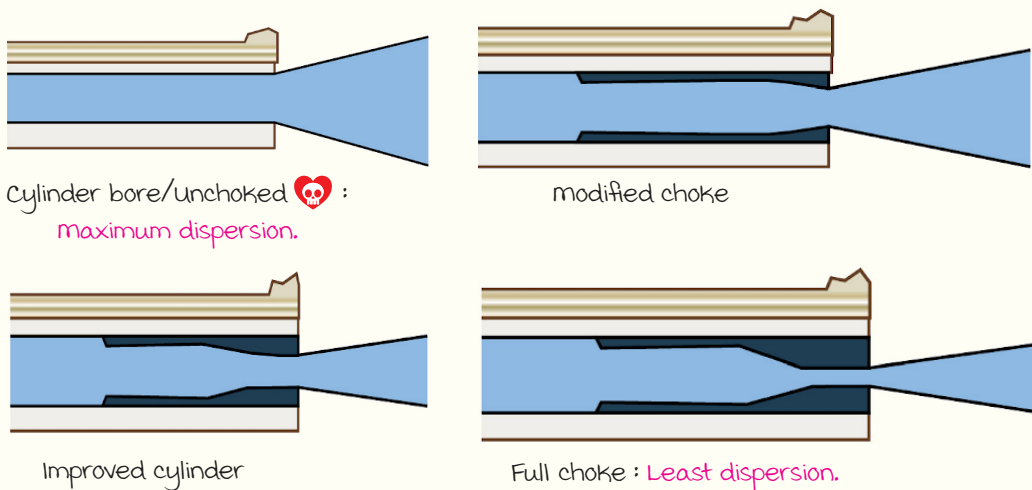
	Rifled gun	Smooth-bored gun/Shotgun
mechanisms	Rifling spins the bullet ↓ ↑ Stability 	Dispersion of lead shots ↓ Poor precision
Projectile	Bullets	Lead shots

----- Active space -----

Choking :

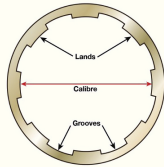
Narrowing the terminal end of the barrel of smooth-bored guns → ↓ Dispersion.

Grades :



Paradox gun : Smooth bored gun with terminal rifling.

Internal Diameter of Firearms :

	Caliber	Gauge
Type of gun	Rifled guns	Smooth-bored guns
measurement	Between 2 diametrically opposite lands 	No. of spherical lead balls made from 1 pound of lead that precisely fit the barrel

----- Active space -----

Ammunition

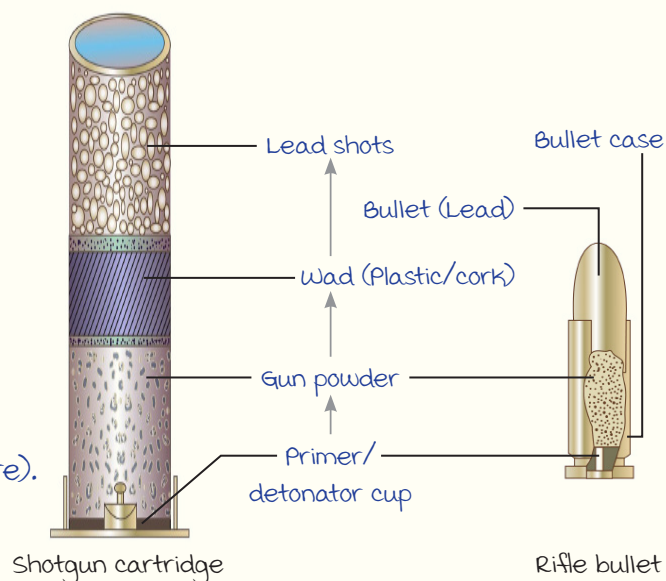
00:05:57

Structure :

Shotgun cartridge :

Wad :

- Travels around 2 - 5 meters.
- Produces **minor bruises**.
- Functions :
 - Lubrication of the barrel.
 - Separation : B/w gun powder & lead shots.
 - Obturation (Seals air pressure).



Rifle bullet :

Wad : **Absent**

(Single bullet directly placed over gun powder).

Primer : (mnemonic : **BLAST**)

- **B**arium nitrate.
- **L**ead peroxide.
- **A**ntimony sulphide.
- **S**typhnate (Lead).
- **T**etrazine.

Types of Gunpowder :

	Black gunpowder 	Semi-smokeless	Smokeless gunpowder
Energy produced	Low (1gm → 3 - 4 litres energy)	-	High (1gm → 12 - 13 litres energy)
Smoke	High	-	Low
Components	• Potassium nitrate (75%) • Charcoal (15%) • Sulphur (10%) 	Black gunpowder (80%) + Smokeless gunpowder (20%) 	• Nitrocellulose • Nitroglycerine • Nitroguanidine 

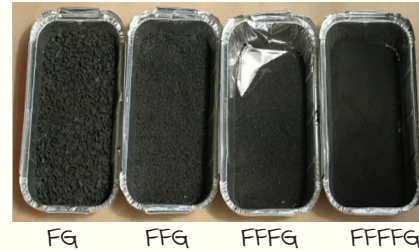
Types of smokeless gunpowder :

- Single base : Nitrocellulose.
- Double base : Nitrocellulose + nitroglycerine.
- Triple base : Nitrocellulose + nitroguanidine + nitroglycerine.

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Fineness of gunpowder :

$FG < FFG < FFFG < FFFFG$
 $\uparrow$ Fineness of gunpowder $\rightarrow \uparrow$ In ability to burn



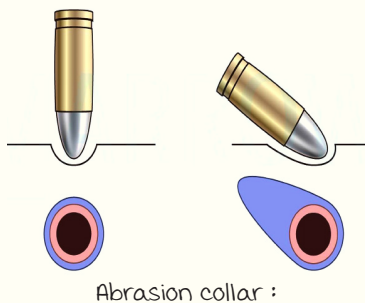
Discharges from a Gun :

In order of emission from the gun :

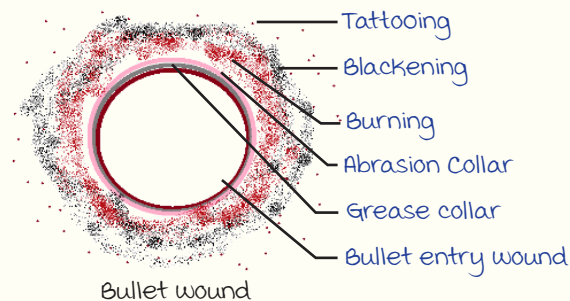
S. No	Discharge	effect
1	Flame	Burns & charring of skin/Scorching
2	Smoke	Blackening/Smudging : Can be wiped with a wet cloth.
3	unburnt gun powder	Tattooing/Peppering/Stippling (Ante-mortem phenomenon) : Can't be wiped with a wet cloth.
4	Bullet 	• Entry wound • Grease collar/Bullet wipe/Dirt collar (Inner collar) : Diagnostic of entry wound • Abrasion collar (Outer collar) : D/t gyroscopic action of bullet

Range :

Discharge	Rifle	Shotgun
Flame	7 cm	15 cm
Smoke	30 cm	45 cm
Gunpowder	50 - 60 cm	60 - 90 cm



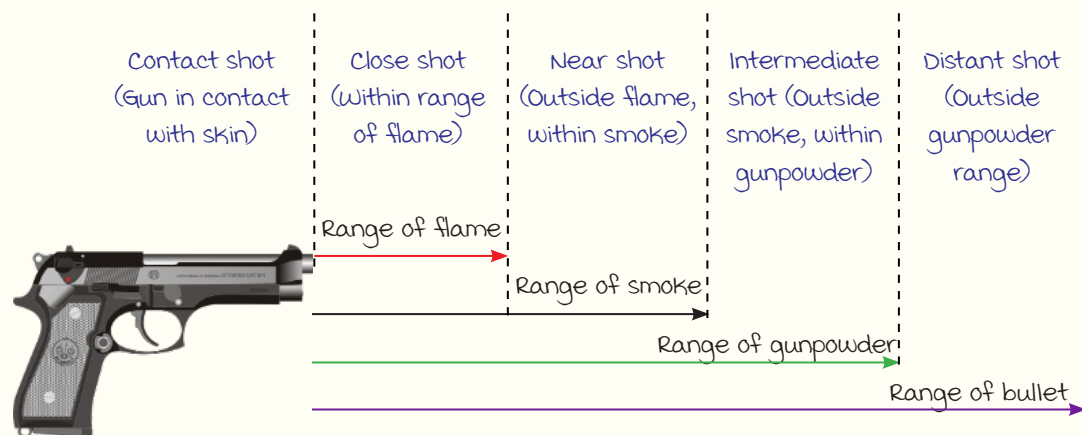
Shape gives information on trajectory of bullet



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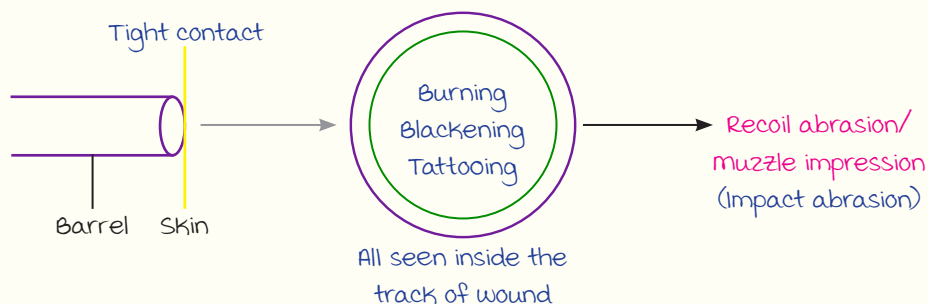
Gun Ranges

00:16:52

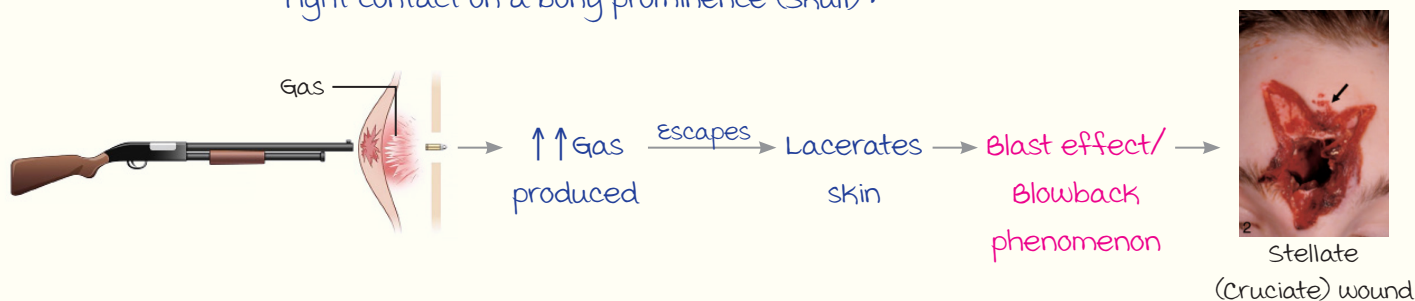
Rifle Gun Ranges :

Contact shot :

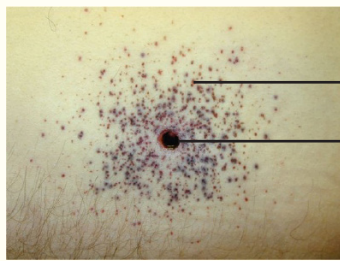
Tight contact :



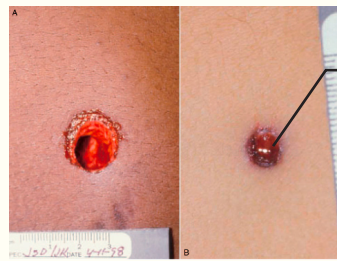
Tight contact on a bony prominence (Skull) :

Wound Characteristics in other Rifle Shots :

Characteristics	Close shot	Near shot	Intermediate shot	Distant shot 🦠
Burning	⊕	⊖	⊖	⊖
Blackening	⊕	⊕	⊖	⊖
Tattooing	⊕	⊕	⊕	⊖
Bullet wound, grease collar & abrasion collar	⊕	⊕	⊕	⊕



Intermediate shot



Distant shot

----- Active space -----

Tattooing
Entry wound

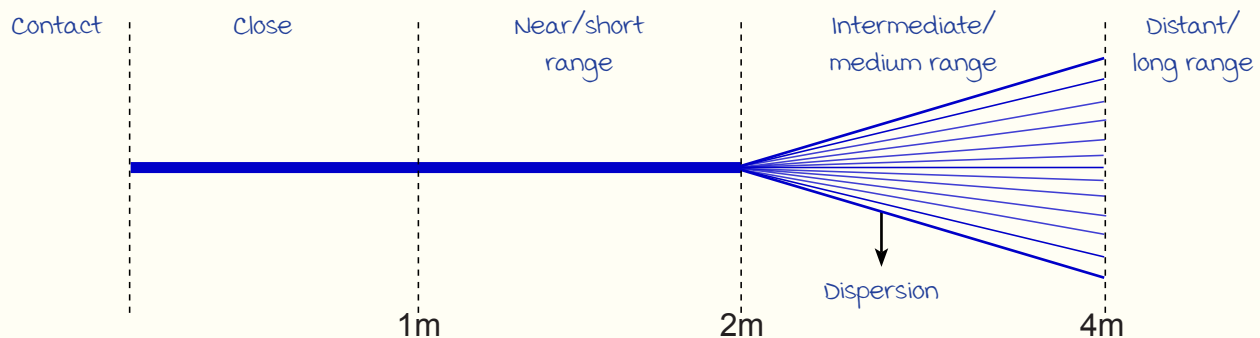
Entry wound
+ grease collar
+ abrasion collar

Entry vs. Exit Wound :

	Entry wound	Exit wound
margins	Inverted	Everted
Burning, blackening, tattooing	Present	Absent
Grease collar & abrasion collar	Present	Absent
Features	Cherry red skin	more bleeding and spattering (Tissue protrusion)

Entry wound with everted margin (Exception) → Contact shot on bony prominence.

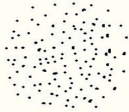
Short Gun Ranges :



Burning /Blackening/Tattooing → Seen within 90 cm.

Range	Wound	Characteristics
Contact		- Stellate or cruciate margin - Burning/Blackening/Tattooing seen inside wound
Close		- All lead shots enter as a single mass - Tattooing ± blackening ± burning : ⊕
Near		- Central single hole with lead shots as single mass - Scalloping of margins ⊕
Intermediate		Dispersion has started - Central hole with satellite holes Satellite holes : Independent pellet holes seen around the wound

----- Active space -----

Range	Wound	Characteristics
Distant		• Complete dispersion • Every pellet seen as independent hole



Satellite holes

Intermediate shot



Distant shot

Atypical dispersion of lead shots :

Incomplete dispersion

Balling/Welding of shots :

- Conversion of shots into compact mass.
- Oval large entry wound.
- Several small circular puncture wounds surrounding.
- mimics use of 2 weapons → Rifle & shotgun.

Erratic dispersion

Billiard ball ricochet effect :

- Ricochet of pellets from an intermediate object.
- Wound mimics distant shot
→ In reality close range shot.

Bullet Fingerprinting & Atypical Bullets

00:30:01

Bullet fingerprinting :

Identifying gun from the bullet markings.

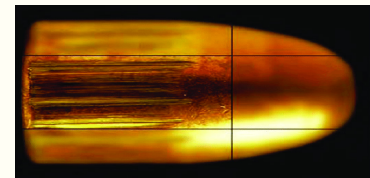
markings on a Gun :

1. Primary marking : macroscopic markings.

- marks produced on a bullet by rifling in the barrel.
- Give information about make/model of gun.



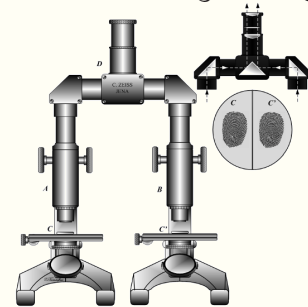
Primary marking



Secondary marking

2. Secondary marking/Fingerprint of a gun : microscopic findings.

- Produced by irregularities of the barrel (Individual characteristics : Vary from gun to gun).
- Unique to each gun.



Comparison microscope :
Compare test bullet (New bullet fired from suspected gun) & crime bullet

Note :Handling of bullet :

- Rubber - tipped tweezers.
(Toothed forceps cause additional markings).
- Gloved hands.

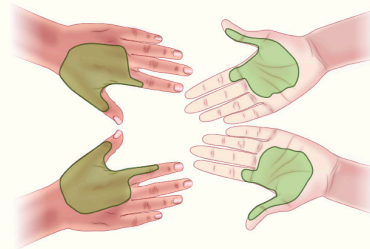


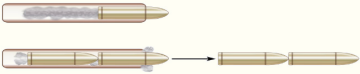
Rubber tipped tweezers

----- Active space -----

Gunshot Residue (GSR) Test :

- To find out if accused has used the gun.
- GSR test can be done by : (mnemonic → **HANDS**).
 - **H**arrison Gilroy test.
 - **A**tomistic absorption spectrometry
 - **N**eutron activation analysis.
 - **D**ermal nitrate test.
 - **S**em - EDXA (**S**pecific test).

Areas of maximum GSR deposition :
Swabs collected & testedAtypical Bullets :

Type of bullet	Features	Image
Tandem/ Piggy-back	2 bullets coming out back to back	
Semi-jacketed/ Dum-dum/ Expanding	Tip of lead core is exposed to outside → mushrooms on entering skin & cause greater damage	
Yawning	Irregular bullet path	
Tumbling	Bullet rotates end to end	
Frangible	Fragments on impact	
Souvenir	• Retained bullet inside the body (Surrounded by fibrosis) • Chronic complication : Plumbism	

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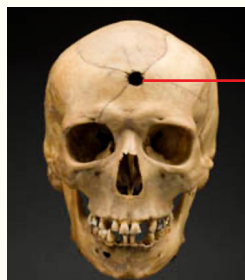
Type of bullet	Features	Image
Ricochet	- Bullet stops spinning on hitting intermediate surface → No abrasion collar - No effects of flame, smoke, or gunpowder on the entry wound - Atypical shape entry wound (keyhole shaped)	
Tracer	Luminous metal at the base → Glows → Can trace the path of the bullet (Phosphorous)	
Kennedy phenomenon	iatrogenic alteration of gunshot wound ↓ Difficulty in range determination.	
Poisoned	Carries curare/ricin/aflatoxin	

Gunshot Wound on the Skull

00:37:42

Bevelling : Differentiates entry wound from exit wound.

- Bevelling in inner table → Entry wound.
- Bevelling in outer table → Exit wound.



Entry wound

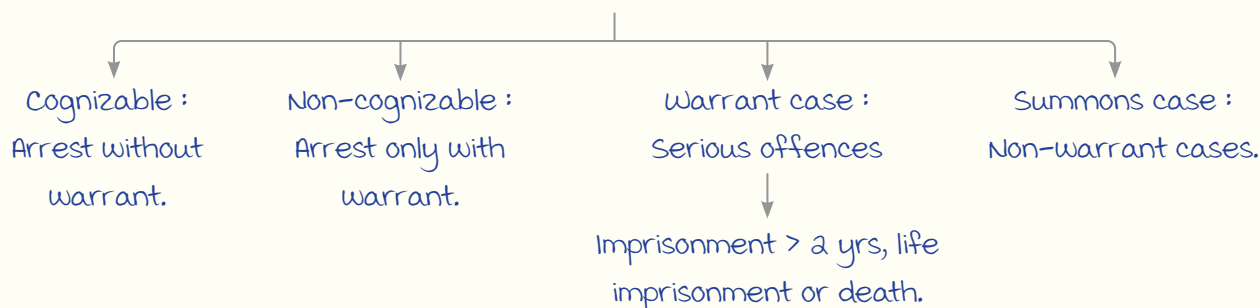


Exit wound

MEDICAL JURISPRUDENCE

----- Active space -----

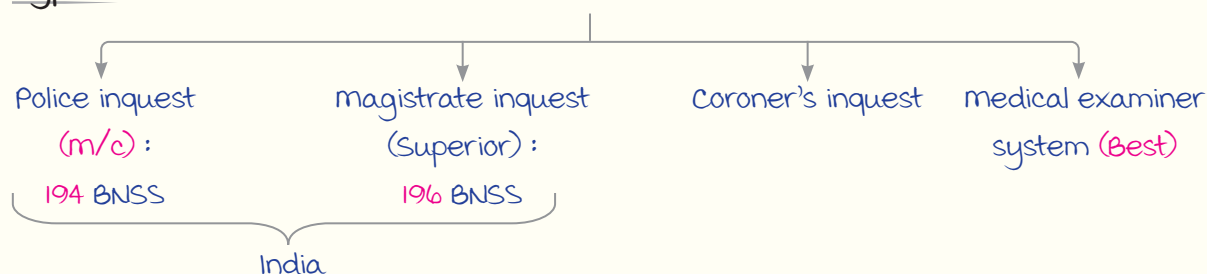
Offences :



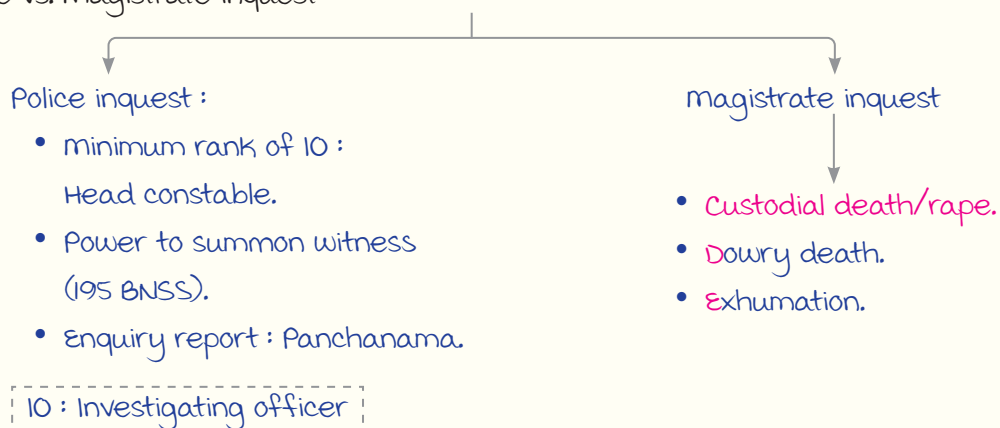
Inquest

00:00:56

Types :



Police vs. magistrate inquest :

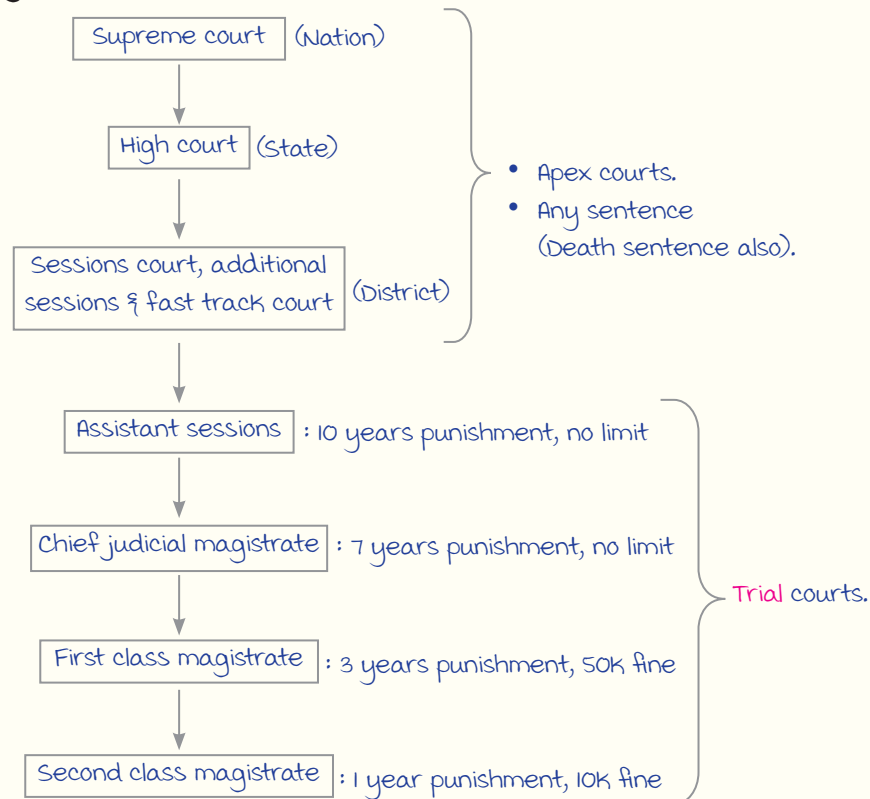


Custodial death : Death while under police/judicial custody.

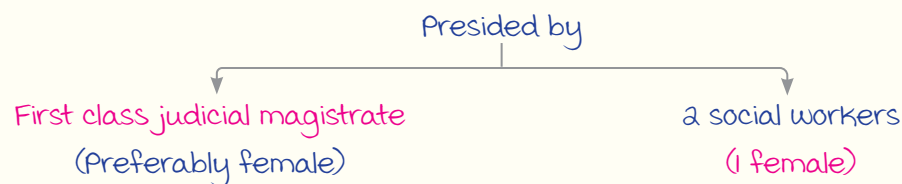
----- Active space -----

Indian Legal System

00:03:43

Hierarchy :

- Commutation of punishment : **Higher** court.
- **Lowest court** to give death sentence : **District** courts.
- Lowest court to confirm/commute/cancel death sentence : High court.
- Death sentence by supreme court can be commuted by : **President/Governor**.
- Death sentence in **pregnant female** → High court $\xrightarrow{\text{Commute}}$ Death to life imprisonment : **456 BNSS**.

Juvenile Court :

Evidence & Witness

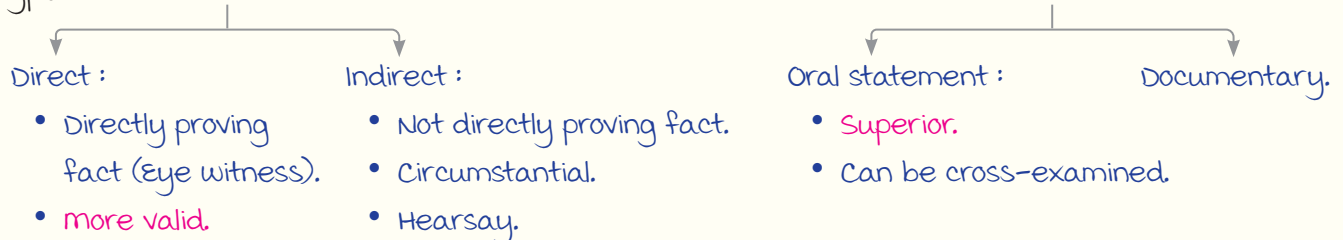
00:07:44

----- Active space -----

Evidence :

Statement/fact produced in court to prove/disprove something.

Types :



Witness :

	Common/Occurrence/Facts witness	Expert witness
Eligibility	- Any person who has perceived the fact - No age limit	By virtue of : - Knowledge/skill - Experience - Training
First hand knowledge rule 	+	-

Doctor  → Can be both common & expert witness.

Summons

00:10:53

- AKA **Subpoena** (Under penalty for civil/criminal court).
- Legal document** issued by the presiding **officer of the court**.
- Compelling the **attendance** of witness.
- If ignored : **Wilful disobedience** (Punishable).

Types :

- Subpoena **ad testificandum** : Give oral statement.
- Subpoena **duces tecum** : Produce documentation.

Order of attendance in case of multiple summons  :

- Higher court > Lower court.
- Criminal court > Civil court.
- Equal status of courts : 1st one to summon.

Conduct/Diet money (Common & expert) :

- Paid **to witness** for traveling expenses (While serving summon in **civil cases**).
- Paid **by parties** in the civil case.
- Amount fixed by **court**.

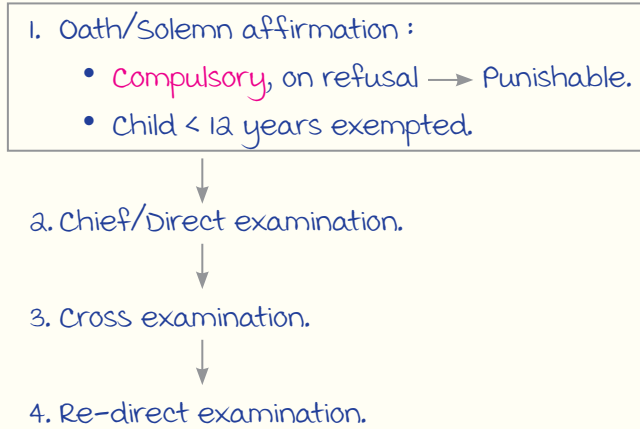
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Court Proceedings

00:13:49

Recording of Evidence in the Court 📌 :

Steps :



Note :

- **Perjury** (False evidence under oath) : 227 BNS
- Punishment for perjury : 229 BNS

Difference in direct vs. cross vs. redirect examination :

	Chief/Direct examination	Cross examination 📌	Redirect examination
Done by	Same side lawyer	Opposite side lawyer	Same side
Prosecution witness	PP	DL	PP
Defense witness	DL	PP	DL
Leading questions 📌	Not permitted	Permitted	Not permitted
Significance	To place all facts before the court	To test accuracy & discredit the witness	To clear any doubts/misinterpretations

Extra powers of a judge :

- Questions can be asked at any stage of trial.
- Can re-call/re-examine witnesses.

- PP : Public prosecutor
- DL : Defence lawyer

Hostile witness 📌 :

- AKA Adverse witness/unfavourable witness.
- Conceals the truth.
- **Contradicting** his previous statement.
- Same side lawyer can ask leading questions.

Dying Declaration vs. Dying Deposition 💀 :

----- Active space -----

	Dying declaration	Dying deposition
Recorded by	• Anyone • Ideally : magistrate	Only magistrate/judge (Superior)
Oath	⊖	⊕
Presence of accused/lawyer	⊖	⊕
Cross examination & leading questions	⊖	
If patient dies	valid	valid
If patient survives	Less value (Corroborative)	valid
India	⊕	⊖

Medical Ethics

00:20:58

medical Ethics vs. Etiquette :

medical ethics :

- Compulsory & punishable.
- Guiding moral principles for :
 - a. Doctor - doctor
 - b. Doctor - patient
 - c. Doctor - state
 Relationships.

medical etiquette :

- Not compulsory.
- Dealing with colleagues.
- Not punishable.

Infamous/Unethical Conduct/Professional misconduct :

Warning notice is given to all doctors informing about :

- Addiction
 - Alcohol
 - Adultery.
 - Conducting criminal Abortion.
 - Association with unqualified person.
 - Advertisements (Inappropriate).
 - Bribery.
- Practicing under influence.
- Covering : Assisting/employing unqualified person.
 - Dichotomy : Fee splitting.
 - Euthanasia : Active is crime.
 - False certificate issue.
 - Gifts from pharmaceutical companies.

Disciplinary action taken by :

- State medical Council (SMC)
 - Ethics & medical Registration Board (EMRB)
- Warning.
 - Remove name from state medical register (Penal erasure) :
 - Temporary.
 - Permanent (Professional death sentence).

----- Active space -----

Appeal :

If punished, one can appeal to higher body :

SMC → EMRB → NMC (National medical Commission).

Professional Secrecy :

- Implied contract.
- B/w doctor & patient to maintain confidentiality.

Privileged communication :

Exceptions for professional secrecy

Notifiable diseases
(E.g. : Communicable diseases, snake bite)

Crime → mandatory Police
intimation (33 BNSS)

Declarations :

Declaration	Guidelines about
Tokyo	Torture
Istanbul protocol	Documentation & investigation of torture victims
Helsinki	Human experimentation
Geneva	medical ethics (modified Hippocratic Oath)
Oslo 	Therapeutic abortion
Sydney	Death declaration

Declaration	Guidelines about
Venice	Terminal illness
Lisbon	Patient rights
Malta	Hunger strike
Washington	Biological weapon

Basic Ethical Principles :

Autonomy : Right of patient
to choose treatment

Beneficence : For the
benefit of patient

Justice : Treat all patients
same, without discrimination

Non-maleficence  :
Do no harm

Medical Negligence/Professional Malpractices

00:28:24

Factors Comprising Negligence :

Duty of care

Dereliction of duty :
Deficiency in duty :
Absence of reasonable skill
or care/wilful negligence.

Direct causation

Damage :

- monetary loss.
- Suffering from pain.
- Disability.
- Death.

Types of Negligence :

----- Active space -----

	Civil	Criminal
Court	Civil/Consumer	Criminal
Parties	Patient vs. Doctor	State vs. Doctor
Burden of proof  (Plaintiff : Person filing case)	Patient	Prosecution
Punishment	monetary compensation	Jail/Fine

Doctrines :

Res ipsa loquitur (Fact speaks for itself) :

- Gross negligent act by doctor.
- No expert opinion needed.
- Burden of proof of innocence → Doctor.
- Applies to both civil & criminal negligence.

Respondeat superior/Captain of ship doctrine :

- Superior is liable for negligent act of the third party.
- Employer is responsible for the negligence of the employee (vicarious liability).
- Only applicable for civil cases.

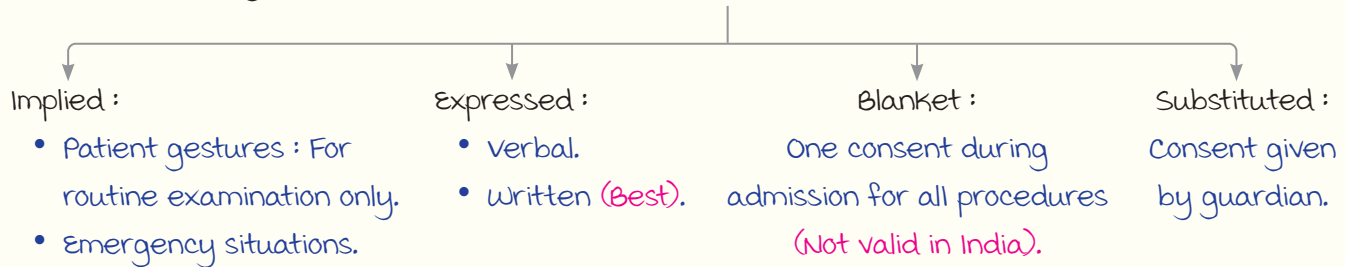
Defences :

- No duty.
- Duty given as per standard protocol : No dereliction of duty.
- Therapeutic misadventure : Accidental damage → Doctor not liable.
- Error of judgement.
- Novus actus interveniens : New unrelated intervening act leading to damage.
- Res indicata (Time limitation period) : Case should be filed within 2 years of knowing.
- Res judicata (Things already decided) : Patient cannot file case again if unsatisfied (Only appeal).
- Contributory negligence : Patient contributed negligence.
 - Partial defense (To reduce monetary compensation).
 - Applicable only for civil cases.
- Product's liability : Poor quality drugs/instruments → manufacturer liable for patient damage.
- Calculated risk doctrine : Doctor → Calculated risk for → Not liable.
patient benefit

Consent

Treatment without consent → Assault.

Types :



Informed Consent :

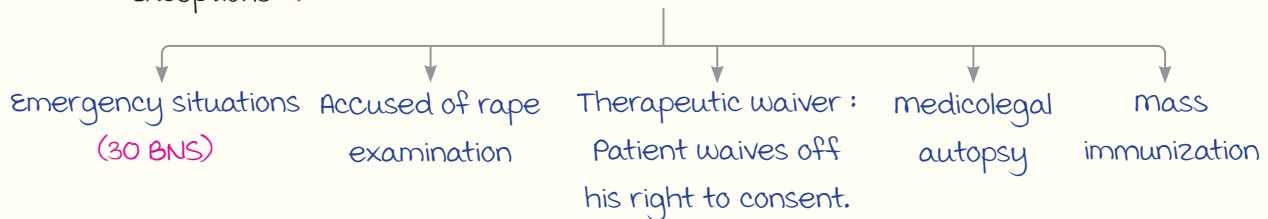
Doctor explains all the benefits & risks of the procedure to obtain consent.

Doctrine of full disclosure : Disclose all the information.

Doctrine of partial disclosure/Therapeutic privilege : Exception for full disclosure.

- Doctor decides information disclosed to patient.
- For patient's benefit.

Exceptions 📛 :



Minimum Age & Validity of Consent 📛 :

Situation	Consent
MTP	> 18 years
major procedure	> 18 years
Physical examination	> 12 years
Child < 12 yrs, insanity, influence, intoxication	Invalid
Children < 12 years (Residential school/Trip)	Loco parentis (Person in charge)

AUTOPSY TECHNIQUES & THANATOLOGY

----- Active space -----

Autopsy : Types

00:00:12

medicolegal vs. Pathological Autopsy 🧠 :

	medicolegal autopsy (m/c)	Pathological autopsy
Type of death	unnatural death	Natural death
Authorization	Investigating officer	Consent of relatives : mandatory
Body handover		Relatives
Examination	Complete autopsy	Partial autopsy

Psychological Autopsy :

- To confirm suicidal death.
- Objective : To assess mental state of the deceased.
- Procedure : Interview family & relatives.

Virtual Autopsy/Virtopsy :

Complete imaging of body (CT/MRI).

Obscure Autopsy vs. Negative Autopsy :

	Obscure autopsy	Negative autopsy
Examination findings	Insignificant	No findings
Opinion on cause of death	After additional tests	Negative (2 - 5% of all autopsies)

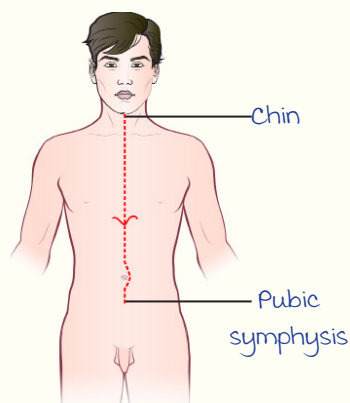
Autopsy : Sequence, Incision, Techniques

00:03:03

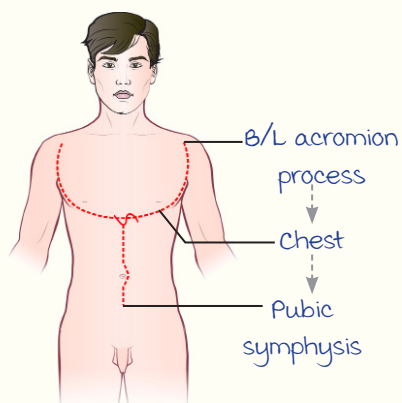
Sequence of Dissection :

Cause	Cavity dissected first	Significance
Poisoning	Cranial	To detect smell of poison
Asphyxial death	Cranial	For bloodless dissection of neck (Dissected last) ↓ Avoid Prinsloo-Gordon artefacts
Pneumothorax	Thorax	To demonstrate air in pleural cavity
Newborn (mnemonic : HAT)	Head → Abdomen → Thorax	Level of diaphragm $\xrightarrow{\text{Respiration}}$ $\begin{cases} \oplus \rightarrow \text{Lower} \\ \ominus \rightarrow \text{Higher} \end{cases}$

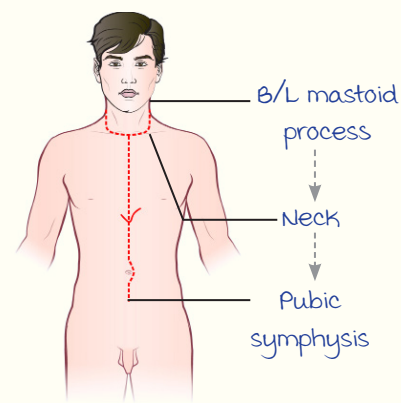
----- Active space ----- Types of Incisions :



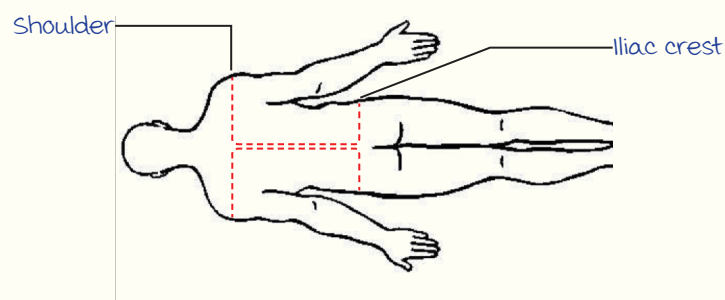
I shaped incision (m/c) :
Simplest incision



Y shaped incision :
In females (Cosmetic purpose)



modified Y shaped incision :
Detailed neck examination



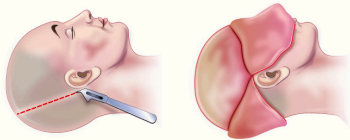
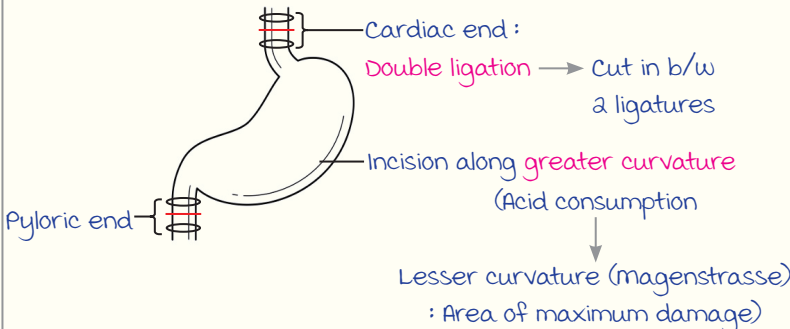
X incision : Custodial deaths

Techniques of Organ Removal 💀 :

	Removal of organ	Features
Virchow's technique	Organ by organ	most common
Letulle's technique	En masse	• Rapid removal • Study inter-organ relationship
Ghon's technique	Blocks : Cervico-thoracic/ abdominal/urogenital block	-
Rokitansky technique	In-situ dissection	In infections (HIV)

Organ Dissection Techniques :

----- Active space -----

	Dissection methods
Scalp	 Bimastroid incision
Skull vault	- Adults → Stryker saw - Fetus → Scissors - Beneke's technique → 2 flaps - Barr's technique (m/c) → 4 flaps
Brain	- Fresh dissection (m/c) - Formalin fixation (Ideal)
Spinal cord	- Not routinely done - Exception : Death d/t : - Whiplash injury - Strychnine poisoning Anterior approach Posterior approach (m/c)
Heart	Inflow outflow method : RA → RV → LA → LV
Stomach	 Cardiac end : Double ligation → Cut in b/w 2 ligatures Incision along greater curvature (Acid consumption) Lesser curvature (Magenstrasse) : Area of maximum damage Pyloric end

Exhumation :

- AKA **disinterment**.
- Lawful digging out of already buried body.

Authorization : magistrate (Section 196 BNSS).

Time limit : No time limit in India.

Preservation of soil samples :

Importance : To differentiate from PM imbibition (m/c : **Arsenic** > Endrin).

Thanatology

00:12:39

- Thanatology : Study of death.
- Taphonomy : Study of post-mortem resorption of body.
- Tripod of life** : Circulation, respiration, brain function.

----- Active space -----

Supravital Period :

- Period between : Clinical/somatic death $\longrightarrow$ molecular death (Cellular death).
stoppage of circulation,
respiration & brain function)
- Significance :
 - Organ harvesting.
 - max. time limit variable for different tissues (E.g. : Cornea $\longrightarrow$ 6 hours).
- Zasko's phenomenon  :
 - mechanical stimulation of muscle $\longrightarrow$ Tendon reaction
(m/c : Quadriceps contraction).
 - seen 2 - 3 hrs after death.
- Electrical excitability of skeletal muscle (+).

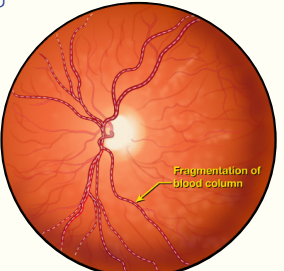
Post-mortem Changes :

Immediate	Early	Late (Decomposition) 
• Loss of voluntary movement • Irreversible stoppage of circulation & respiration	• Eye changes • Algor mortis (Cooling) • Livor mortis (Staining) • Rigor mortis (Rigidity)	• Autolysis : Lysosomal enzymes • Putrefaction : - Bacterial enzymes (Lecithinase) - Clostridium welchii

Early Changes

00:16:40

Eye Changes :

	Features	Time since death (TSD)
Retina	Kevorkian sign/cattle trucking sign : Fragmentation of retinal vessels 	Few minutes $\longrightarrow$ 1 hour
Sclera 	Tache noire sclerotica : Triangular opacities on either sides of cornea 	3 - 6 hours

most reliable sample for TSD in decomposed body : vitreous humor ($\uparrow K^+$) > Blood.

Algor mortis/Post-mortem Cooling :

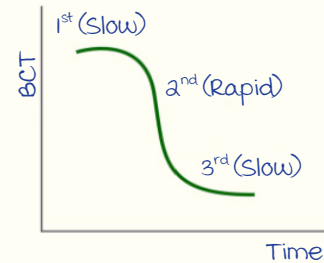
----- Active space -----

↓ Body core temperature (BCT) :

- Instrument : **Thanatometer**/chemical thermometer.
- Site : **Rectum** (m/c except in sodomy cases), subhepatic space, nose, tympanic membrane, lower end of esophagus.

Post-mortem caloricity 🦴 :

- Body remains warm after death.
- Causes : D/t high BCT at time of death.
 - Heat stroke.
 - Pontine hemorrhage.
 - Septicemia.
 - Tetanus/strychnine poisoning.
- Not seen in **burns**.



Sigmoid/inverted S curve :
3 phases of fall of BCT

Livor mortis 🦴 :

- AKA cadaveric or post-mortem lividity/Pm staining/Pm hypostasis/**sugillation**/cogitation.
- Hypostasis of blood.



Supine position :
Back of head, chest,
abdomen, legs



Prone position :
Front of head, chest,
abdomen, legs



Vertical position :
Glove & stocking
distribution

- Hypostasis absent in :
 - Body is in a flowing river.
 - Severe anemia/shock.

Features :

- Onset : **30 mins**.
- Visible : Within 1 - 4 hours.
- Max. : 6 - 12 hours.
- merges with putrefaction colour.

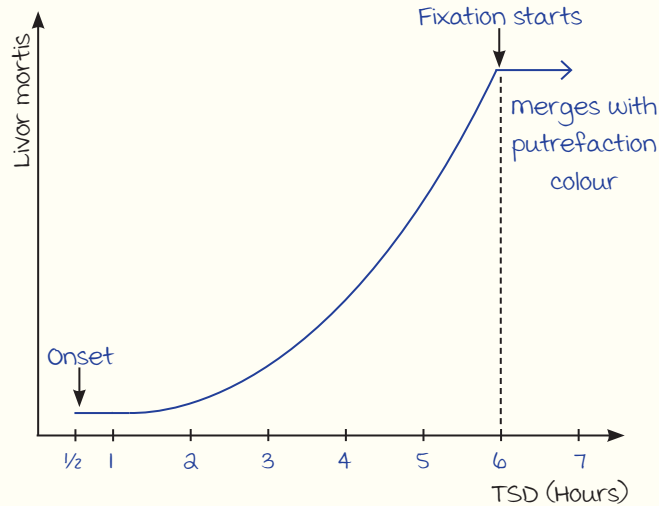


Blanching (+) : Not fixed

----- Active space -----

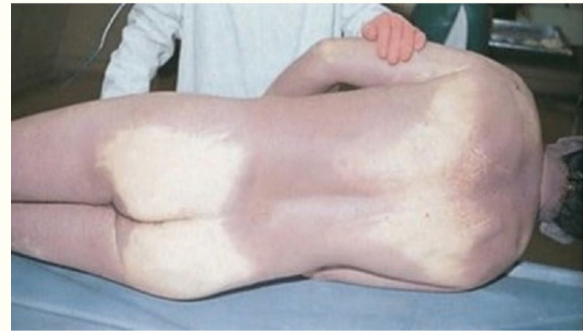
- **Fixation :**

- Happens at 6 - 12 hours.
- **Blanching ⊖** : Fixed.
- movement of body before fixation : Primary lividity → **Secondary lividity.**



- **Contact pallor** : Areas in tight contact → Pallor.
- Colour 💀 :

Colour	Cause
Cherry red	Carbon monoxide
Bright red	Cyanide
Brown	• Phosphorus • Aniline • Nitrate
Black	Opium
Bluish green	Hydrogen sulfide



Contact pallor

Rigor mortis 💀 :

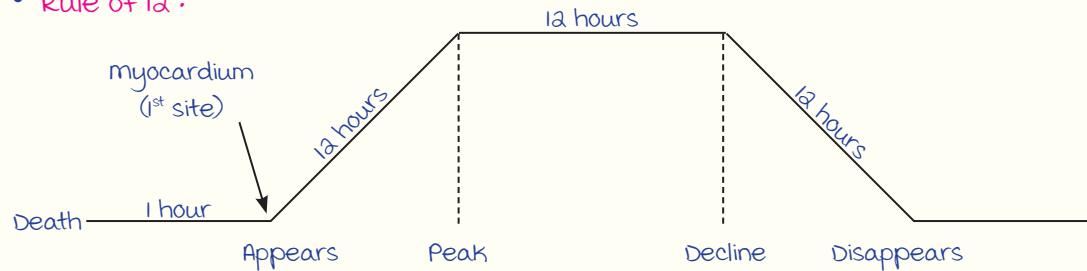
- AKA cadaveric stiffening/rigidity.
- Primary flaccidity $\xrightarrow{1 \text{ hr}}$ muscle stiffness $\xrightarrow[\text{1 - 2 days}]{\text{Proteolysis (Decomposition)}}$ Secondary flaccidity.
(Sphincter relaxation ⊕) (ATP depletion)

↓
Rigor mortis)

- Sequence : Involuntary (1st in myocardium) → voluntary muscles.
- Externally : **Nysten's rule** (Head to toe).

Eyelid → Neck → Jaw → Face → Thorax → Upper limb → Abdomen
 ↓
 Fingers & toes. ← Lower limb

- Rule of 12 :



Cadaveric spasm/instantaneous rigor/cataleptic rigidity :

- No primary relaxation.
- Spasm of groups of voluntary muscles.
- Always **antemortem**.
- Drowning, electrocution, suicidal gunshot.



Cadaveric spasm

Late Changes (Putrefaction)

00:30:51

- Colour change.
- Gas production.
- Tissue liquefaction (5 - 10 days after death).

Colour Change 🦴 :

- 1st site : **Aorta** → Reddish-brown discolouration.
- 1st external site :

Rt. iliac fossa → Greenish discolouration → Time since death :

- Summer : 12 - 18 hrs.
- Winter : 1 - 2 days.

marbling :

- Green > Brown.
- D/t **sulfhemoglobin** deposition.
- TSD : 36 - 72 hours.



marbling

Gas Production :

- **Hydrogen sulphide**.
- TSD : 2 - 3 days.

Abdomen
distension



Gas stiffening

----- Active space -----



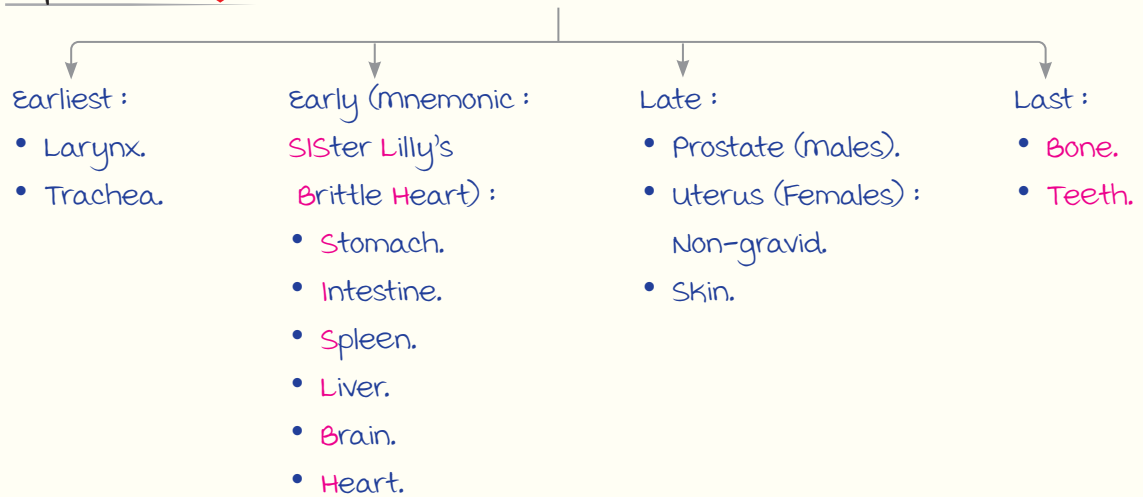
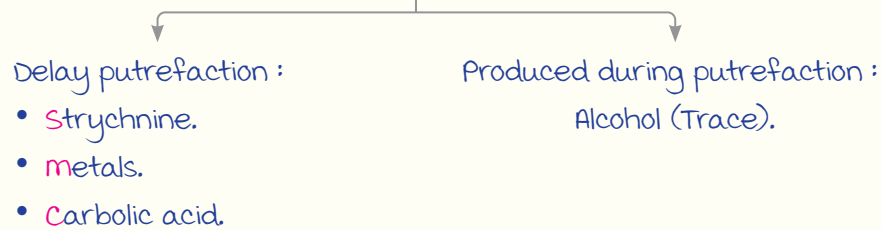
Pm blisters

Bleeding from
nostrils

Pm purge

Note :

	Pm blister	Antemortem blister
Content	Gas	Inflammatory fluid
Base	Pale	Reddish

Liquefaction 💀 :Poisons :Casper's Dictum :

Rate of putrefaction : Air (Fastest) > Water > Earth (Slowest).

1

2

8

Entomology :

- Study of insects.
- Identify :
 - Time since death (Stage of insect).
 - Poison.
 - Place of disposal (Based on insect species).

modified Forms of Putrefaction :

----- Active space -----

	Adipocere/saponification/grave wax 🦴	mummification
Conditions required	• Warm climate • moisture • Clostridium welchii (Anaerobe) • Still air	• Hot & dry climate • Free air movement • Arsenic & antimony poisoning
mechanism	Fat (Starts in subcutaneous layer) Hydrolysis & hydrogenation ↓ Fatty acids ↓ Greyish white waxy substance	Drying and dehydration ↓ Shrunk body (Dry, brown & brittle)
Smell	Sweet rancid odour	Odourless
Duration	3 weeks - 3 months	3 - 12 months
Fetus	⊕ (> 7 months IUL)	⊕
Image		

Note :**Embalming :** Thanatopraxia.

- Embalming fluid : **Ethanol** not used.
- Injection technique : High pressure, low volume.
- Ideal time : < 6 hours of death.
- Poisoning case :
Embalming prior to autopsy → Punishable (**238 BNS** : Disappearance of evidence).

Viscera Preservation for Chemical Analysis

00:40:53

For poison detection.

Routinely Sent Samples :

- **Blood** (most reliable) : Peripheral vein.
 - Stomach & it's contents
 - Small intestine
- } Both in one bottle.
- Liver (500 gms)
 - Kidney (Half of each)
- } Both in another bottle.

----- Active space -----

Samples Sent for Specific Poisons :

Poison	Specimen
- Strychnine - Gelsemium	Spinal cord (Full length)
Digitalis	Heart
- Organophosphates - Alcohol - morphine	Brain
Pesticides	Adipose tissue
metals	Bone, hair, nail
Gas	Lungs
Alcohol	CSF
CO/cyanide	Spleen

Preservatives :

- Saturated sodium chloride solution (m/c) :

Avoided in :

- Aconite poisoning.
- Corrosives except carbolic acid.

- Rectified spirit (Ideal) :

Avoided in :

- Formalin poisoning.
- Phosphorus poisoning.
- Alcohol poisoning.

Specimen	Preservatives
Blood	Sodium fluoride + potassium oxalate
- Urine - Vitreous - CSF	Sodium fluoride
Virology	50% glycerol

Samples Sent for DNA Analysis 💀 :

- Blood → EDTA preservative.
- Muscle.
- Bone.
- Hair.
- Teeth :
Dental pulp → DNA.

Note :

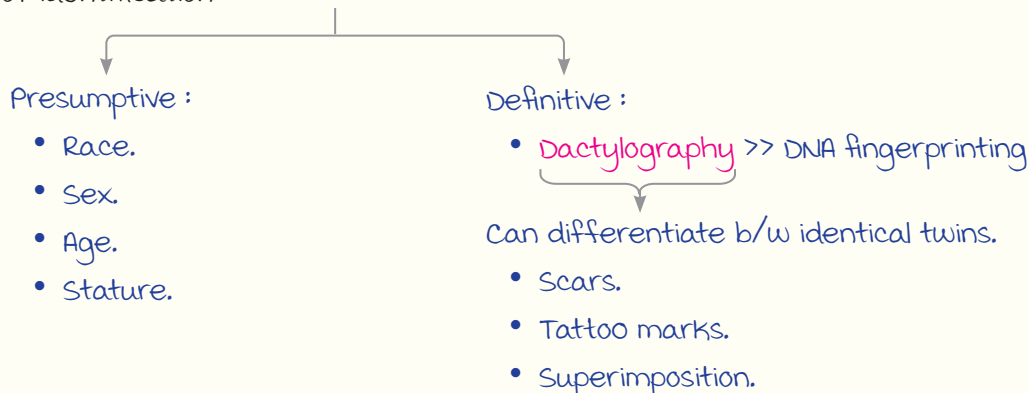
Blood stained cells → Air dried → Sent in paper envelope.

HUMAN IDENTIFICATION

----- Active space -----

Corpus delicti : Body of offense/essence of crime.

Types of identification :



Identification of Race, Sex and Age

00:00:56

Race :

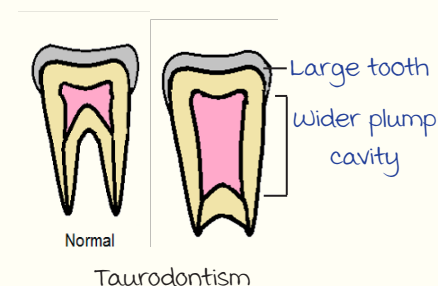
Indices for race determination :

Mnemonic : **BCCI**.

- **B**rachial index : From upper limb.
- **C**ephalic index : From cranium (**Best**).
- **C**ruial index : From lower limb.
- **I**ntermembral index : Compare upper & lower limb.

Cephalic index 🦴 :

- Cephalic index = $\frac{\text{max breadth of skull}}{\text{max length of skull}} \times 100$.
- 70 - 74.9 : Dolicocephalic → Aryans/Africans.
- 75 - 79.9 : mesocephalic → Indian/Chinese/Europeans (mnemonic : **ICE**).
- 80 - 85 : Brachycephalic → Japanese.



Shovel shaped incisors

Dental features :

Race	Negroid	Caucasoid	Mongoloid
Dental features	• ↑ Tooth size • ↑ Cusps	Carabelli's cusp	Mnemonic : PASTE • Pointed canines • Absence of 3rd molar • Shovel incisors 🦴 • Taurodontism 🦴 • Enamel pearl (Premolars)

----- Active space -----

Sex :

Concealed sex :

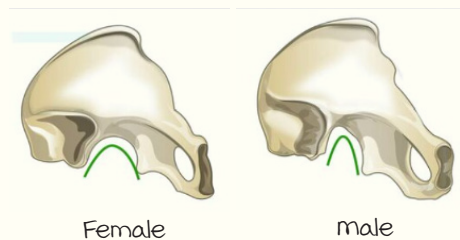
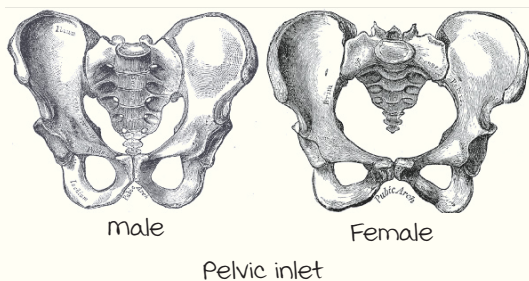
Wearing clothes of opposite sex → Hiding sex for a motive (Revealed O/e).

Note :

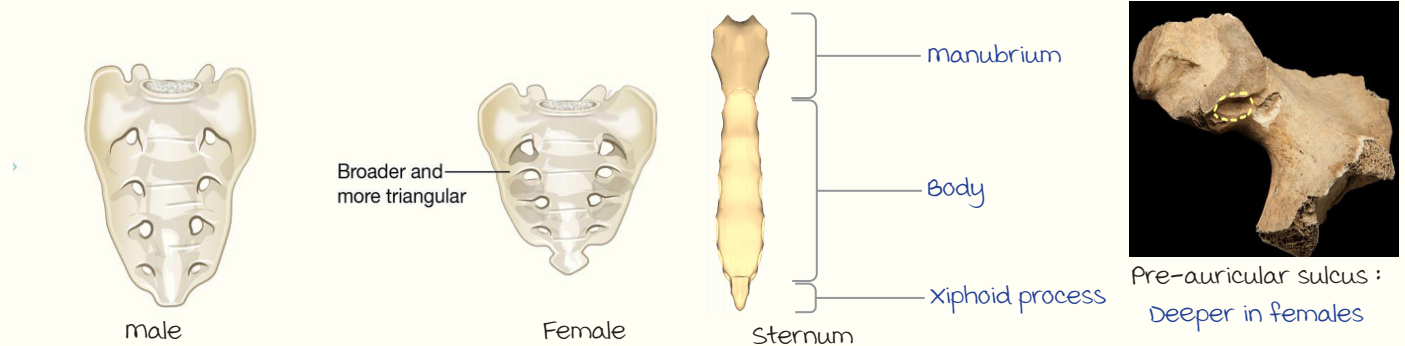
Transvestism : Wearing clothes of opposite sex → Sexual gratification.

Sex determination 💀 :

		male	Female
muscle markings		more prominent	- Less prominent - Exceptions 💀 : - Frontal and parietal eminence (Skull) - Pre-auricular sulcus (Pelvis)
Shape	Skull (Orbits, chin)	Square	Rounded
	Obturator foramen	Large and oval	Small and triangular
Forehead		Sloping	Vertical
Pelvic inlet		Heart shaped	Circular
Acetabulum and sacroiliac articulation		Larger	Smaller
Angle	mandibular	Less obtuse : $< 125^\circ$	more obtuse : $> 125^\circ$
	Subpubic	$< 90^\circ$	$> 90^\circ$
Ischial tuberosity		Inverted	Everted
Greater sciatic notch (Best)		Deep and narrow	Wide and shallow
Higher index		Corporobasal index : $\frac{\text{Breadth of body of } S_1}{\text{Breadth of base of sacrum}} \times 100$	- Sciatic index (Best) - Ischiopubic/Washburn index - Sternal index
Sacrum	Shape	Long and narrow	Short and wide
	Promontory	more prominent	Less prominent
Sternum	Sternal length (manubrium + body) : Ashley's rule of 149	$> 149 \text{ mm}$	$< 149 \text{ mm}$
	Hyrtil's law	Body $>$ 2x length of manubrium	Body $<$ 2x length of manubrium



----- Active space -----

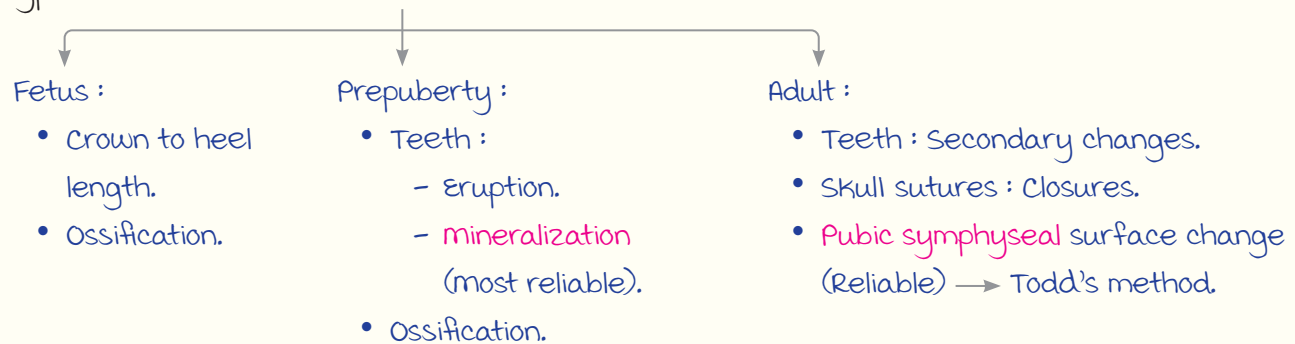


Accuracy of sex determination from bones/Krogman's accuracy :

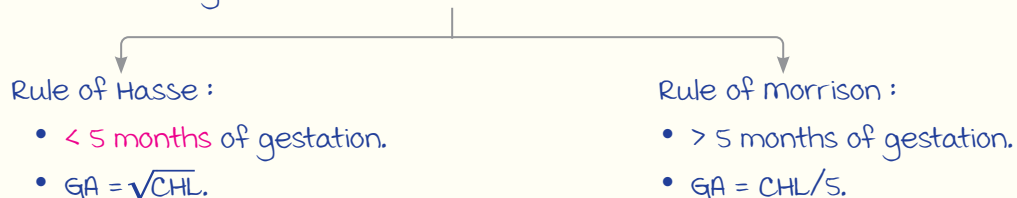
- Pelvis : 95% (Best), even in children/fetus
- Skull: 90 - 92%.
- Long bones : 80%.
- Pelvis + skull : 98%.
- Pelvis + long bones : 98%.
- Complete set : 100%.

Age Estimation :

Types :



Crown to heel length (CHL) :



CHL = Crown rump length (CRL) × 1.5.

----- Active space -----

Ossification Centres

00:17:31

Ankle joint :

Centre	Age of appearance
Calcaneum	5 th month IUL (Before viability)
Talus	7 th month IUL
Femur (Lower end)	36 weeks IUL
Tibia (Upper end)	38 weeks IUL
Cuboid	At birth



mandible :
2 halves fuse at 1 - 2 yrs

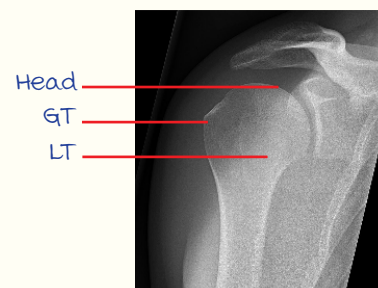


medial end of clavicle : 18 - 22 yrs

- Appear at : 18 - 19 yrs
- Fuses at : 21 - 22 yrs

Humerus :

	Age of appearance	Age of fusion
Head	1 yr	17 - 18 yrs
Greater tubercle	3 yrs	
Lesser tubercle	5 yrs	
Tip of acromion	14 - 15 yrs	



> 17 yrs

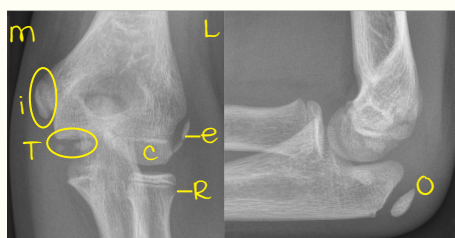
Elbow joint : mnemonic CRITOE.

	Age of appearance	Age of fusion
Capitulum	1 yr	16 - 17 yrs
Radius head	5 yrs	
medial epicondyle (i)	6 yrs	
Trochlea	9 yrs	
Tip of olecranon	9 yrs	
Lateral epicondyle (e)	11 yrs	



> 16 yrs

(All centres appeared
& fused)



- All 6 centres appeared.
- Not fused.
- 11 - 17 years.

Wrist joint :

- Best way to find if adult (> 18 years).

	Age of appearance	Age of fusion
Lower end of radius	2 yrs	18 - 19 yrs
Lower end of ulna	5 yrs	17 - 18 yrs



> 18 y (Lower end of radius fused)



3 yrs - 4 yrs

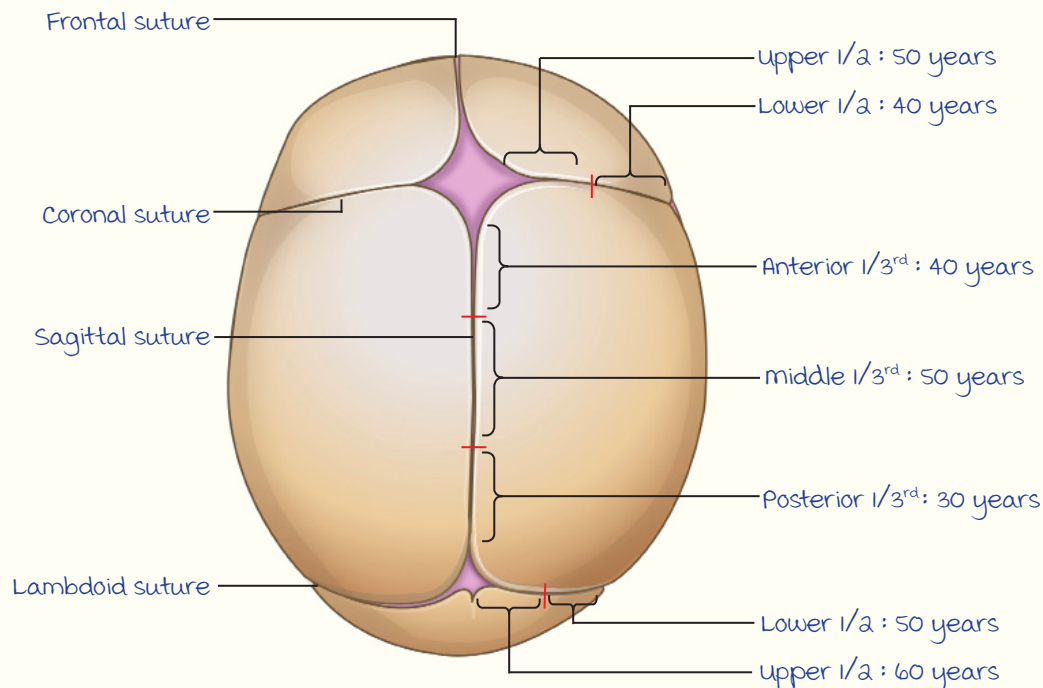
Capitate, hamate, triquetrum
radial end appeared

- Order of appearance of carpal bones :

Capitate → Hamate → Triquetrum → Lunate → Scaphoid → Trapezium/trapezoid → Pisiform
(2 months) (3 m - 1 yr) (3 y) (4 y) (5 y) (Twin bones; 5 - 6 y) (9 - 12 y)

Skull :

- Posterior fontanelle (Lambda) : 3 - 6 months.
- Anterior fontanelle (Bregma) : 18 months.
- metopic suture : 2 - 4 years.



- Endocranial suture (more reliable) > Ectocranial suture.

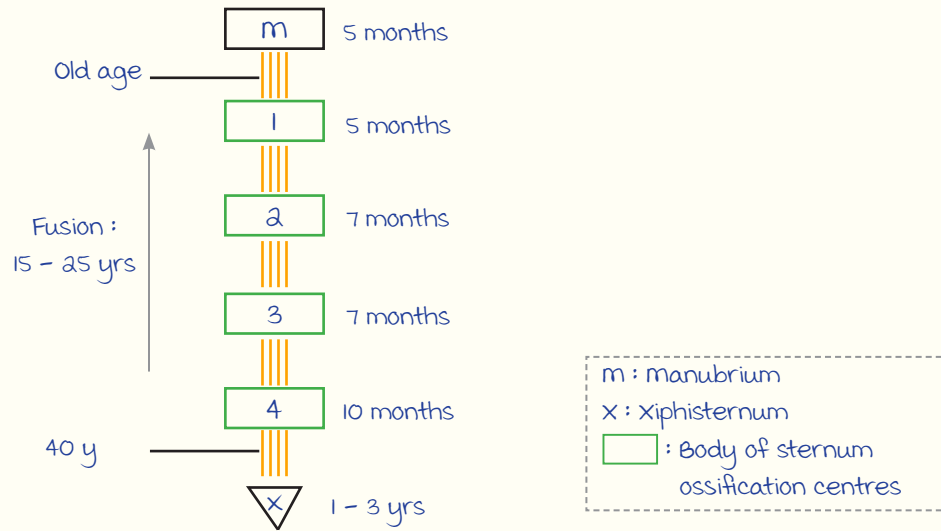
Fuses 5 - 10 years earlier.

- Spheno-occipital suture (Basiphenoid & basiocciput) : 18 - 22 years 🧠.

----- Active space -----

Sacrum  : 22 - 25 years .

Sternum :



Note :

Child under guardianship of court → major at 21 years.

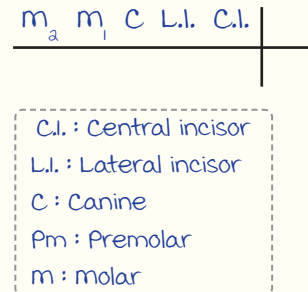
Dentition

00:30:10

Types of Teeth :

Primary/milk teeth/deciduous teeth :

- Temporary : 20.
- Order of eruption : mnemonic **ImCm**
 - **I** : 6 months (First : Lower C.I.)
 - **m_l** : 12 months.
 - **C** : 18 months.
 - **m_a** : 24 months.



Secondary :

- Permanent : 32.
 - Order of eruption :
 - m_l : 6 years (First).
 - C.I. : 7 years.
 - L.I. : 8 years.
 - Pm_l : 9 years.
 - Pm_a : 10 years.
 - C : 11 years.
 - m_a : 12 - 14 years
 - m₃ : 17 - 25 years (wisdom tooth)
- Time gap for spacing of jaw.
Failure : Impacted tooth (m₃ : m/c).



Successional teeth (20) : Ones replacing temporary set.

Super added teeth (12) : Additional ones (All permanent molars).

----- Active space -----

Period of mixed Dentition :

- Period with both temporary & permanent teeth.
- 6 - 11 years.
- Total number constant = 24.

methods of Estimation :

Gustafson's method : After 25 years.

Mnemonic : **APSRTC**.

- Attrition (wear and tear).
- Parodontosis (Gum changes).
- Secondary dentin (2nd reliable)
- Root resorption
- Transparency of root (most reliable)
- Cementum apposition

Anterior → Posterior : Reliability ↓
(C.I. most, m₃ least reliable)

} microscopic changes.

Stack's method : From the height and weight of the tooth.

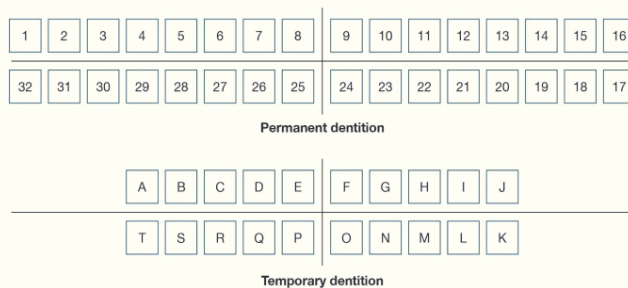
Boyde's method :

- Counting microscopic incremental lines in teeth.
- First line (Neonatal line) : Day 2 or 3 (Sign of live birth).
- Every day, one line.

Dental Charting :

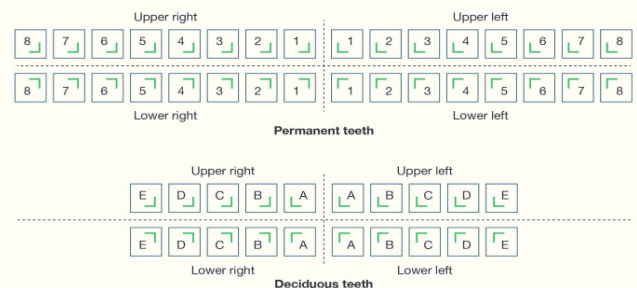
Universal method : Oldest method.

- Continuous numbering 1 - 32.
- Starting right upper and clockwise.



Palmer's notation :

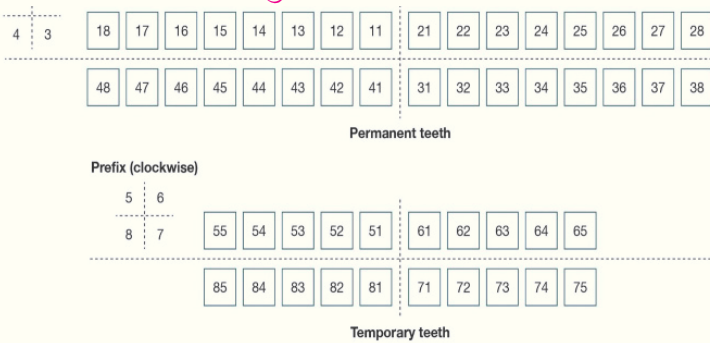
- Count medial → lateral.
- With symbol for quadrants.



----- Active space -----

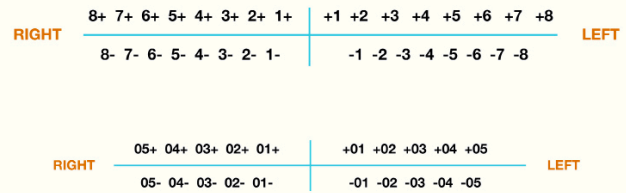
FDI method :

- 2 digit system.
- Permanent teeth prefix : 1, 2, 3, 4 each for specific quadrant in clockwise.
- Temporary teeth prefix : 5, 6, 7, 8.
- Counting teeth 1 - 8, medial → lateral.
- Worldwide mostly used.



Haderup's notation :

- + : maxillary teeth (Upper jaw).
- - : mandibular teeth (Lower jaw).



Stature

00:42:37

Estimation :

1. Regression formulae :

Karl-Pearson formula (Commonly used).

2. Percentile :

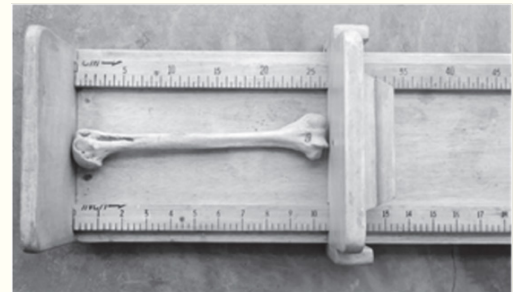
- Femur → 27% of height (Best).
- Tibia → 22%.
- Humerus → 20%.
- Spine → 35%.

3. multiplication factor 🦴 :

Length of bone × multiplication factor = Height.

mnemonic : FeTHUR.

- Femur : 3.7
- Tibia : 4.5
- Humerus : 5.3
- Ulna : 6.1
- Radius : 6.5



Hepburn osteometric board :
used to measure length of long bones

Note :

----- Active space -----

Bertillon's method : Oldest method for identification.

- Anthropometric measurements (11 key body dimensions).
- Descriptive information (E.g. : Eye colour, hair colour).
- Photographs.
- Personal details.

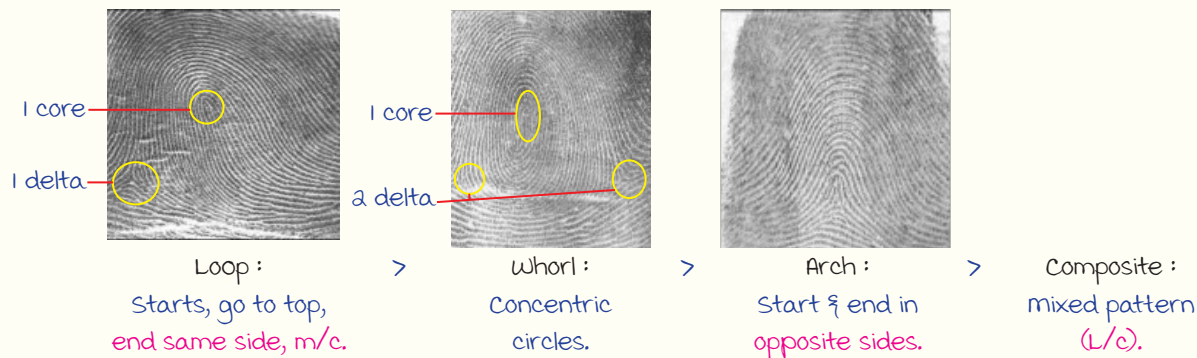
Dactylography and Other Methods

00:44:49

Dactylography :

- Galton's system.
- Develop in intrauterine life :
 - Starts : 12 - 16th week (3 - 4 m).
 - Completed : 20 - 24th week (5 - 6 m).

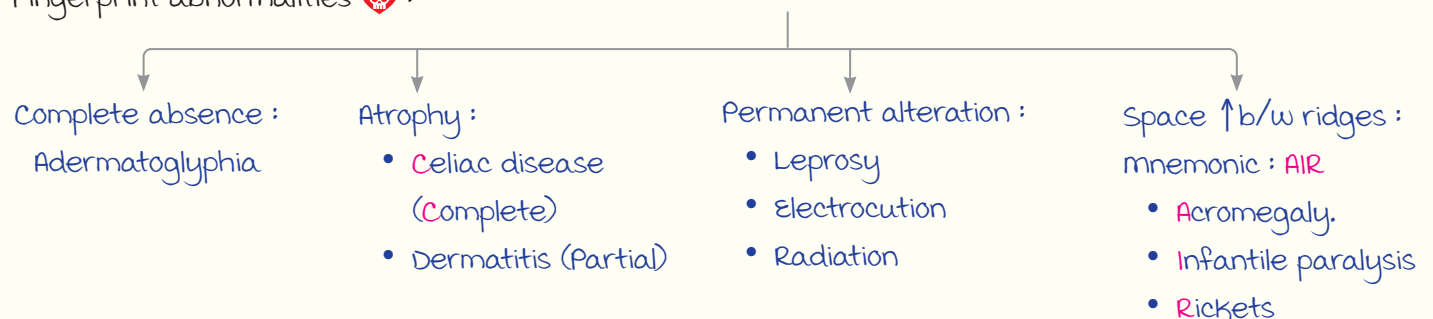
Patterns :



- Core and delta :

	Core	Delta
Pattern	Recurring point	Triradiate
Arch	0	0
Loop	1	1
Whorl	1	2

Fingerprint abnormalities 🦴 :



----- Active space -----

Comparison of fingerprints :

- Patterns not used.
- Compare :
 - **minutiae** (Ridges) → Ridgeology.
 - microscopic pattern from edge of ridges → Edgeoscopy.

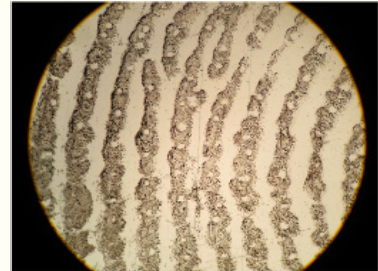
Other methods :

Poroscopy :

Arrangement of **sweat** pores :**Locard's poroscopy.**

Cheiloscopy :

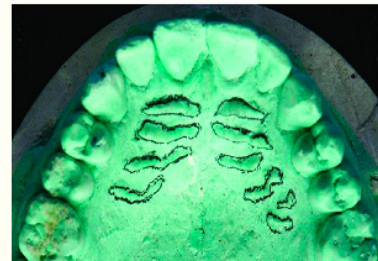
Suzuki classification used → Study of lip prints.



Rugoscopy (Study of rugae)/palatoscopy :

Anterior hard palate is evaluated.

Rugae — Primary (5 - 10 mm).
 — Secondary (3 - 5 mm).
 — Tertiary (< 3 mm).



Aging of scar :

- Angry, red scar → 5 - 6 days.
- Pale, tender → 2 weeks.
- White, soft, ↓tenderness → 2 - 6 months.
- Non-tender and firm → After 6 months.

Tattoo marks :

Faint tattoo visualized by :

- magnifying glass.
- Infrared photography.
- uv light.
- Regional LN examination (**Pigment**).

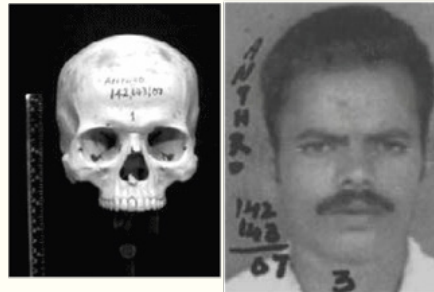
Podogram : Study of footprints.

Superimposition :

- matching the skull & antemortem photograph of missing person.
- Look and match :
 - Inner, outer canthus.
 - Nasion.
- **Negative value tests** : Helps with exclusion.

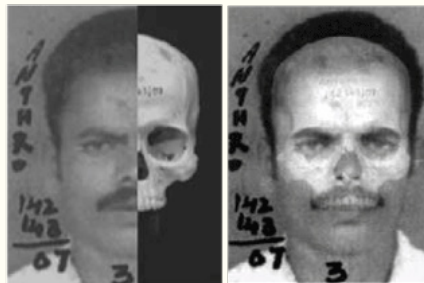
----- Active space -----

Skull-Photograph Superimposition



1. (a) Skull

1.(b) Photograph

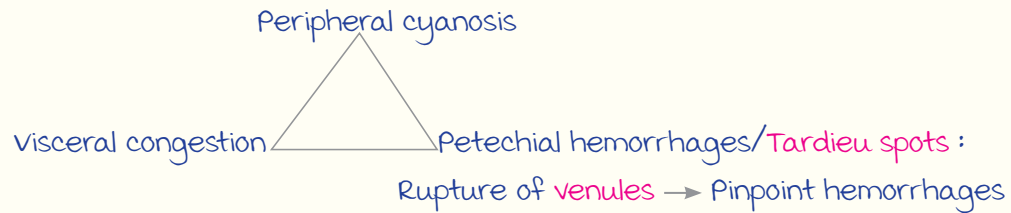


1.(c) Wiped image

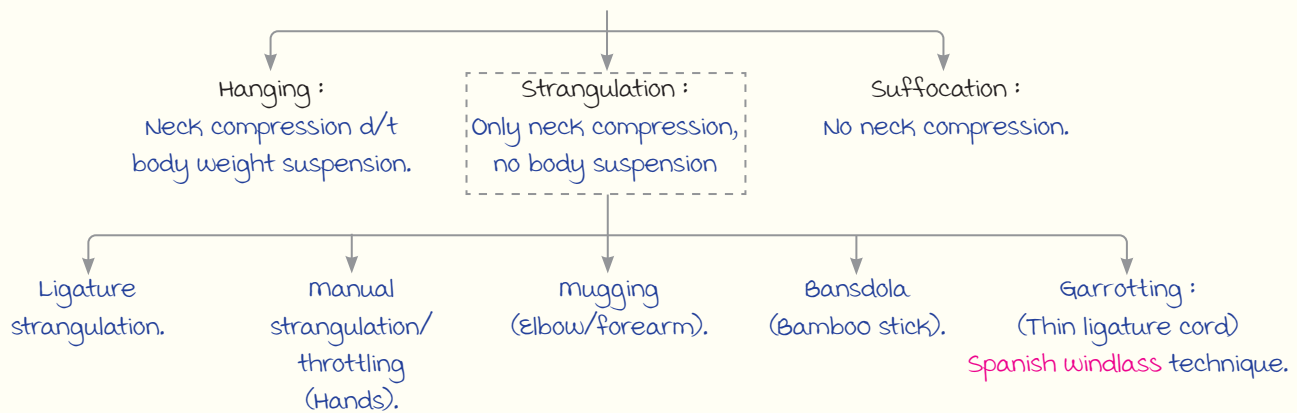
1.(d) Mixed image

ASPHYXIAL DEATHS

Asphyxial triad (Non-specific) :

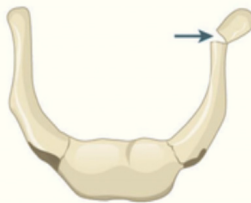
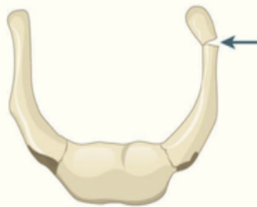


Types Based on mechanism :



Hyoid Bone Fractures :

1. Adduction # (m/c) :
 - Inward compression.
 - Seen in throttling.
2. Abduction #/AP compression # :
 - Outward compression.
 - Seen in hanging.
3. Side to side compression # :
 - One end displaced outwards & other inwards.
 - Seen in hanging.



Hanging

00:04:21

Types :

- Based on knot position →
- Typical hanging :
Knot in occiput.
 - Atypical hanging :
Knot anywhere other than occiput (Judicial hanging).

Based on suspension :

----- Active space -----

Complete hanging	Partial hanging (Incomplete hanging)
whole body suspended	Partial body suspended
No body part touches ground	Body part is touching the ground (E.g. : Leg, knee)
Constricting force : whole weight of the body	Constricting force : Partial weight of the body
Faster death	Slower death (Suicidal)

Autopsy Findings :

Order of autopsy :

- Cranium : Opened first → To avoid Prinsloo-Gordon artefact.
- Neck : Opened last → To facilitate bloodless dissection of neck.

Hypostasis : Glove & stocking pattern.

- Implies body was in vertical position.
- Does not imply hanging.

Face :

- Dribbling of saliva.
- La facie sympathique.



Hanging victim :
Dribbling of saliva

Dribbling of saliva 🦴 :

- Surest sign of antemortem hanging.
- Opposite to knot side.

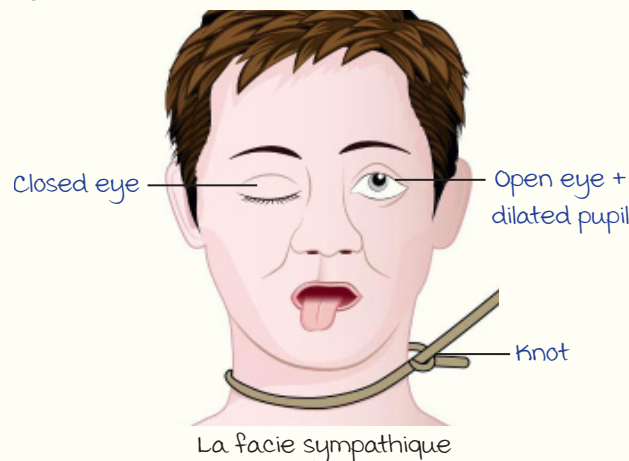
La facie sympathique 🦴 :

- Pressure on cervical sympathetic chain



1/L opening of eyelids +
1/L pupillary dilatation.

- Sign of antemortem hanging.



Ligature mark :

- Features :
 - Oblique.
 - Incomplete.
 - Above thyroid cartilage.
- Transverse ligature mark : Seen in
 - Partial hanging (D/t low point of suspension).
 - Slip knot with running noose.

----- Active space -----



Ligature mark : Oblique, incomplete, above thyroid



Suspension
peak



Pale, dark brown,
parchment-like

Ligature mark : Pressure abrasion

Internal findings :

1. Hyoid bone # :

- Seen in 15 - 20% of hanging cases.
- > 40 years.

2. Carotid artery : Amussat sign (Transverse intimal tear).

3. Vertebra :

- C₂ fracture (Hangman's #) → Cause of death in judicial hanging.
- Simon's hemorrhage (Intervertebral disc hemorrhage).

manner of Hanging :

Suicidal (m/c) :

m/c method of suicide :

Hanging > Poisoning.

Homicidal :

Lynching/

extrajudicial hanging :
Done by a mob.

Accidental :

Sexual asphyxia :

- AKA Asphyxiophilia/hypoxiphillia/ Kotzwarism.
- Asphyxia → Sexual gratification.



Lynching



Sexual asphyxia

Strangulation

00:12:41

----- Active space -----

Ligature Strangulation :

Ligature mark :

- Transverse.
- Complete.
- **Below** thyroid cartilage.
- underlying skin shows extensive extravasation of blood.
- Bruising around ligature mark ⊕.
- Hyoid/cricoid # : ⊖.

Throttling :

Homicidal.

External findings :

- Nail marks.
- **Six penny bruises.**

Internal findings :

- Extensive soft tissue contusion.
- Adduction # of hyoid.
- **Cricoid cartilage #.**



Strangulation ligature mark



Six penny bruises



Tissue contusion in throttling



mugging



Garroting



Spanish windlass
(Garroting variant)



Bansdola
(Bamboo sticks used)

Suffocation

00:15:13

Smothering 💀 :

Closure of mouth & nostrils together :

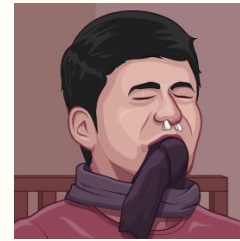
1. Accidental.
2. Homicidal :
 - Intentionally smothering with pillow/hands.
 - If hands are used :
 - **Perioral injuries** (Abrasion/bruises).
 - Lip injuries.



----- Active space -----

Gagging :

Thrusting cloth/pad into mouth → Obstruction of pharynx
↓
Asphyxia.



Gagging

Choking 📛 :

mechanism :

Entry of foreign body into airway
↓
Respiratory distress (Air hunger, gasping)
↓
Asphyxia.

management : **Heimlich maneuver.**

Cafe coronary syndrome :

- Entry of food bolus into airway → Sudden collapse & death.
- mimics myocardial infarction.
- Risk factors : Intoxication.
- Cause of death : **vagal inhibition** of heart.



Heimlich maneuver

Traumatic & Positional Asphyxia :

Traumatic asphyxia/crush asphyxia/Perthes syndrome :

- D/t heavy weight on person's chest.
- **masque ecchymotique** : Pale chest with cyanosed face.
- Burking (Homicidal) : Smothering + Traumatic asphyxia.
- Overlaying (Accidental)

Restriction of chest movements

Asphyxia

Positional asphyxia 📛 :

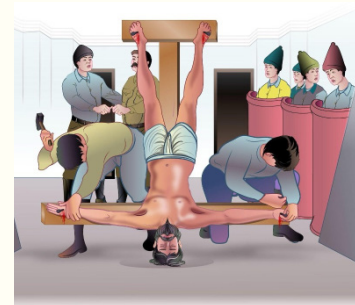
- D/t abnormal position.
- Types :
 - a. **Jack knife.**
 - b. Inverted crucifixion.



masque ecchymotique



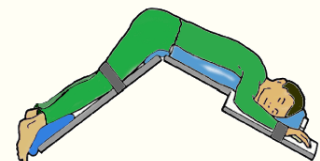
Overlaying



Inverted crucifixion



Burking



Jack knife position

----- Active space -----

- Features :
 - Fine.
 - Tenacious.
 - Persistent (Even after wiping).
- **Absence of froth :**
 - Dry drowning.
 - Hydrocution.
 - Unconscious person.



Cadaveric spasm



Frothing

Non-specific findings :

1. Cutis anserina
 2. Washerwoman's hand/feet : Imbibition of water into skin.
- Features :
 - Wrinkling.
 - Peeling of cuticle.
 - Bleaching.
 - Soddening.
 - Indicative of only **time since immersion**.
 - Starts in fingers (3 - 4hrs) $\xrightarrow{24 \text{ hrs}}$ whole hand.



Washerwoman's foot

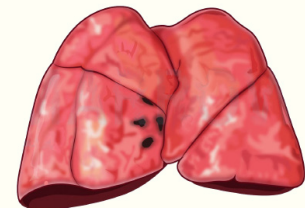


Cutis anserina

Lung changes :

Emphysema aquosum : Conscious drowning.

- Voluminous
- Crepitant
- Ballooning out.
- **mud particles** in lower airway.
- **Palttauf hemorrhages**.
- Frothing (+).



Palttauf hemorrhage

Oedema aquosum : Unconscious drowning.

- Lung edema (+++).
- ↓ Frothing.

Other findings :

1. Presence of water in
 - Stomach/Intestine.
 - Sinuses.
 - middle ear.
2. middle ear hemorrhages.
3. Shrunken spleen : **Sabinsky's sign**.

Tests for Drowning :

----- Active space -----

Gettler's test :

Principle : Compare Cl^- concentration of blood in right and left heart chambers.

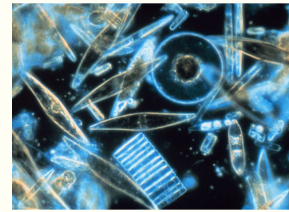
Inference :

- Normal : Left = Right (Cl^- concentration)
- Wet drowning : > 25% difference between left & right side.

Cl^- concentration	Effect	Drowning type
$\text{Rt} > \text{Lt}$	Hemodilution	Freshwater
$\text{Lt} > \text{Rt}$	Hemoconcentration	Salt water

Diatom test 🦴 :

- Features of diatom :
 - unicellular algae.
 - Outer wall has silica :
Resistant to acid/alkali & putrefaction.
- microscopy :
 - Observation : Presence of diatoms in spleen,
bone marrow (Best site), brain & liver.
 - Inference :
Suggestive of antemortem drowning (Diatoms entered d/t intact circulation).



Diatoms : microscopy

Note :

	Palttauf hemorrhages	Petechial hemorrhages
Specificity	Specific to drowning	Non-specific asphyxial sign
Location	Only lung	multiple organs
Cause	Rupture of alveoli	Rupture of venules

SEXUAL JURISPRUDENCE AND TRACE EVIDENCE

Terminology :

male	Impotency : Inability to achieve & maintain penile erection (m/c cause : Anxiety)	Satyriasis 🦴 : Excessive sexual desire	Impotence quoad hanc : Impotence towards one particular woman (Psychological)
Female	Frigidity : Sexual coldness	Nymphomania : Excessive sexual desire	-

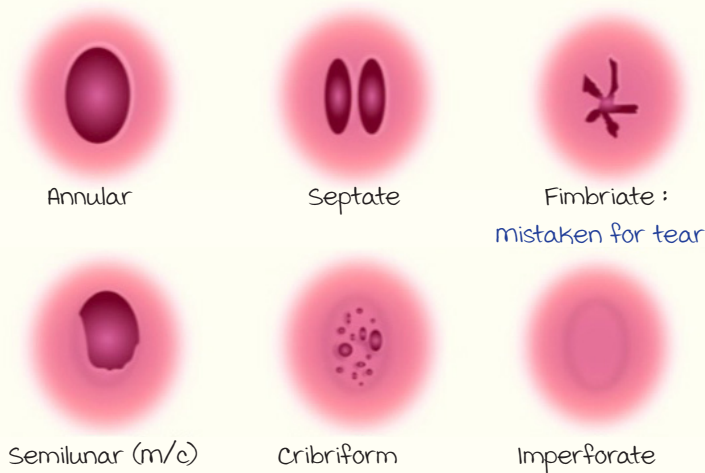
Fecundation ab extra : Conception in a female without penile penetration.

Hymen & Terminologies Related to Pregnancy

00:01:10

Hymen 🦴 :

Types :



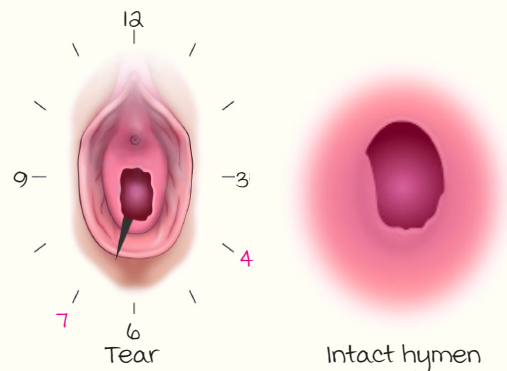
Hymen tear :

- Posterolateral : Penile penetration.
- Anterior : Digital/foreign body.

Intact hymen :

Intact even after intercourse :

- False virgin : Too elastic/loose/thick.
- Female child : Deep seated.



Terminologies Related to Pregnancy :

----- Active space -----

Phantom/Spurious pregnancy/Pseudocyesis :

- Female believes she's pregnant.
- Presentation :
 - Amenorrhea.
 - Abdominal distension.
 - morning sickness.
- No confirmatory signs.
- On USG :
 - No fetal parts/heart sound.
 - Empty uterus.
- Rx : Counseling.

Superfecundation vs. Superfetation  :

	Superfecundation	Superfetation (Rare)
Fertilisation of	2 ova	2 ova
Ovulatory cycle	Same cycle	Different
Acts of coitus	2	2
Partner	1 or 2	
Fetal growth	Equal/Fetus compressus/ Fetus papyraceous	Unequal

Supposititious/Fictitious/Substituted/Forged child :

Female may feign pregnancy & claim some other child as her own.

Atavism : Child resembling grandparents.

Posthumous  : Child born after the death of father.

Affiliation case :

- Female alleges a man to be the father of her child.
- For fixing paternity.

Lochia :

- Sign of recent delivery.
- Lochia rubra → Lochia serosa → Lochia alba.

Surrogacy Regulation Act

00:06:16

Surrogacy :

Surrogate woman bears and hands over the child post-delivery to an intending couple.

Types :

	Altruistic surrogacy	Commercial surrogacy
Features	• No monetary compensation to the surrogate mother • Only medical expenses and insurance covered	monetary compensation given to the surrogate mother
Permitted under surrogacy act	Yes	No

----- Active space -----

Eligibility Criteria :

Intending couple	Intending woman	Surrogate woman
• married (> 5 years) • Age limit : - male : 26 - 55 years - Female : 23 - 50 years • No surviving child • Essentiality & eligibility certificate (+)	• widow or divorcee • Age : 35 - 45 years	• married • Age : 23 - 35 years • At least one child of her own • Eligible only once in lifetime (max 3 attempts) • Health insurance (3 years) : Provided by intending couple

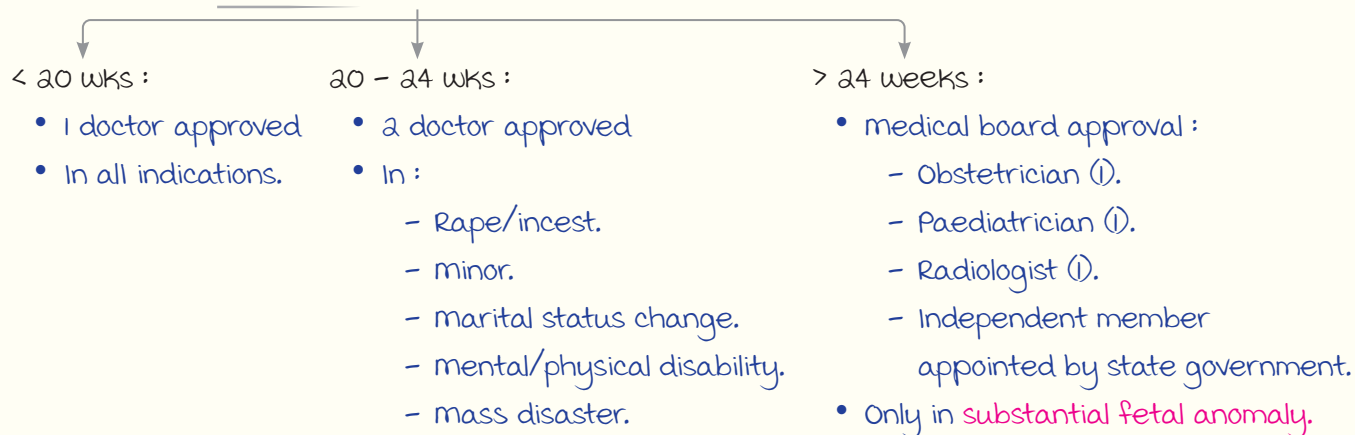
- unmarried person/couple
 - Foreign nationals
 - At least one gamete from biological parent.
- } Not eligible.

Medical Termination of Pregnancy (Amendment) Act 2021

00:08:31

Indications :

- Eugenic : Fetal abnormalities.
- Therapeutic : maternal disease.
- Social : Failure of contraception.
- Humanitarian : Rape.

Duration :Consent for MTP :

- min age > 18 yrs.
- < 18 yrs/mentally ill : Consent of guardian.

MTP Records :

- Preserve it for 5 years.
- To be forwarded monthly to district CMO.
- Strictly confidential : violation punishable.

Criminal Abortion/Voluntary miscarriage :

- 88 BNS : With consent of female.
- 89 BNS : Without consent of female.
- 90 BNS : With/without consent, resulting in death of female.

Infant Death

00:12:08

----- Active space -----

- under 103 BNS : Amounts to murder.
- Killing of :
 - Infant (< 1 yr) : Infanticide.
 - Fetus by parents : Filicide.

Maceration :

- Sign of intrauterine death.
- Aseptic autolysis.
- Skin reddening/slippage (Earliest sign : 12 hrs).
- Hypermobility joints.

Radiological Signs :

- Robert sign : Gas shadow in great vessels (12 hrs).
- Ball's sign : Hyperflexion of spine.
- Spalding sign : Seen in 4-7 days, Overriding of cranial vault bones.

Postmortem Examination :

	method	Not respired	Respired
Diaphragm	Check level	3 rd - 4 th rib	6 th - 7 th rib
Fodere's test/Static test	Lung weight measured	30 g	60 g
Ploucquet's test	Lung weight/Baby weight (measured)	1 : 70	1 : 35
Wreden's test	middle ear content checked	Gelatinous tissue	Clear
Breslau's second life test	Stomach is perforated in water	No air bubble	Air bubble
Raygat's/Hydrostatic test/ Lung floatation test	Lung cut in pieces and put in water	Sink	Floats
	Specific gravity measured (Residual air)	1.040	0.940

Note :

1. Cry of baby in the vagina : vagitus vaginalis.
2. Cry of baby in the uterus : vagitus uterinus.

Child Abuse

00:17:19

Battered Baby Syndrome/Caffey Syndrome :

- AKA Caffey-Kempe syndrome/Non-accidental violence.
- Child abuse → Repetitive physical injuries.
- By parents or guardian.

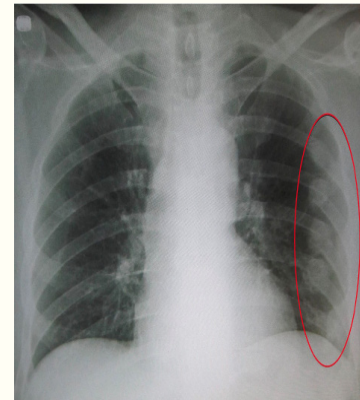
----- Active space -----

Features :

- Discrepancy b/w nature of injuries & history given.
- Delay b/w : Injury and medical attention.
- Multiple, repetitive injuries of **various ages**.

Common injuries :

- **Laceration of oral mucosa + frenulum of lower lip** (Characteristic).
- Egg shell fracture (Skull).
- Bucket handle fracture (Long bones).
- Knob fracture/String of bead appearance (CXR).



String of beads appearance :
CXR

Munchausen's Syndrome by Proxy :

- Type of abuse.
- Parent **fabricates** illness in child → Brings child to hospital (To gain attention).

Sudden Infant Death Syndrome (SIDS)/Cot Death/Crib Death

00:19:36

- Sudden unexpected death of a seemingly healthy infant.
- Death remains unexplained even after thorough investigations.

Probable causes :

1. **Prolonged sleep apnea**.
2. Respiratory infection.
3. Hypersensitivity to cow's milk.

Autopsy findings :

- Usually **negative**.
- Constant finding : **Petechial hemorrhages** on heart, lung, thymus.

Sexual Offences

00:20:26

Rape 🩸 :

- 63 BNS : Definition of rape.
- **Statutory rape** : victim age < 18 yrs (With/without consent).
- 64 BNS : Punishment of rape → 10 years to life imprisonment.
- 64 (a) BNS : Punishment of custodial rape.
- 67 BNS : Punishment for forceful sexual intercourse with wife during separation.
- 70 BNS : Punishment for gang rape.
- 366 (a) BNSS : **In-camera trial** (Closed court proceedings).
- **397 BNSS** :
 - All doctors must treat victims of rape/vitriolage/POCSO :
 - Immediately.
 - Free of cost.
 - Police intimation mandatory.
 - Violation : Punishable under **200 BNS**.

medical examination :

----- Active space -----

	Accused	Victim
Section	52 BNSS	184 BNSS
Consent	Not mandatory	• mandatory • Patient denies examination → Informed refusal taken → Treatment for : • Injuries • STD • Pregnancy
Tests	Lugol's iodine test : • Detects vaginal epithelial cells • Swab from penis shaft is exposed to iodine vapour → Brown color. • upto 4 days	• Toluidine blue test : Recent microinjuries. • motile sperm (6 - 12 hr) : Recent rape.

Preservation of samples (vaginal swab) :

- Sperm : 3 days.
- Semen : 4 days.

Unnatural Sexual Offences :

Not punishable if b/w 2 consensual adults.

Incest	Lesbianism/Tribadism/ Sapphism	Sodomy/Greek love/ Buggery	Buccal coitus/Oral sex/ Sin of Gomorrah
Sex b/w family members : • Oedipus complex : mother & son • Electra complex : Father & daughter • Pharaon complex : Brother & sister	B/w females Active partner : Dyke Passive partner : Femme	• Penile anal sex • If an adult male with : Elderly : Gerontophilia Child (Catamite) : Pederasty.	Oral stimulation of : • Penis : Fellatio. • Vagina : Cunnilingus.

Bestiality : Intercourse with lower animals.

----- Active space -----

Sexual Perversions :

Sexual gratification without intercourse.

Paraphilia	mode of sexual gratification
Sadism/Allogolagnia	By inflicting pain/physical cruelty
Masochism/passive algolagnia	By receiving painful stimulus from opposite partner
Lust murder	By killing victim
Bondage	Sadist + masochist
Transvestism/Eonism	By wearing dress of opposite sex.
Exhibitionism	- Recurrent, intense, sexual urge to expose one's genital to an unsuspecting stranger. - Punishable under 296 BNS : - Streaking : Publicly runs nude - mooning : Shows gluteal region - Flashing : Suddenly undressing & exposing private parts
Voyeurism/Scotophilia	- Watching private acts of unsuspecting female (Peeping tom) - Punishable under 77 BNS
Fetishism	With inanimate objects/non-living things
Frotteurism	- Rubbing genitalia against non-consenting person in public - Punishable under 75 BNS
Scatologia	Talking obscenity
Klismaphilia	Taking enema
Urophilia/Undinism/Urolagnia	Sight/smell/thought of urine
Coprophilia	Sight/smell/thought of feces.
Ecouterism	Listening to sounds a/w sexual intercourse.
Necrophilia	Intercourse with dead body
Necrophagia	Eating dead body
Troilism	Observing wife having sexual intercourse with another man
Narratophilia	By telling obscene stories

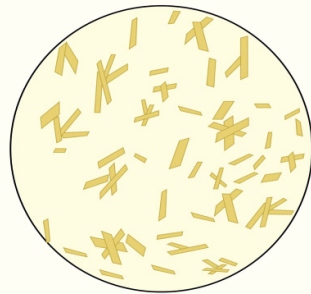
Evidence Collection

00:33:16

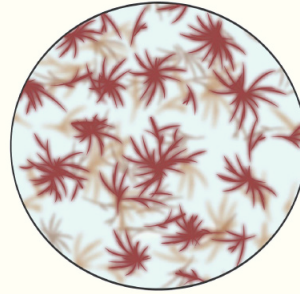
Test for Blood Stains :

Presumptive tests		Confirmatory test
Color test (Reagent : H_2O_2)	Luminescent test	
Benzidine test/Orthotoluidine test : Blue color $\rightarrow$ $\oplus$	To detect blood even if it has been washed off : - Luminol spray. - uv light.	1. microscopy 2. microchemical test : - Teichmann's test : Brown rhombic crystal (Haemin). - Takayama test : Pink feathery crystal (Haemochromogen) 3. Absorption spectrometry (most reliable) : Recent/old stains
Phenolphthalein's/Kastle Meyer test : Pink color $\rightarrow$ $\oplus$		

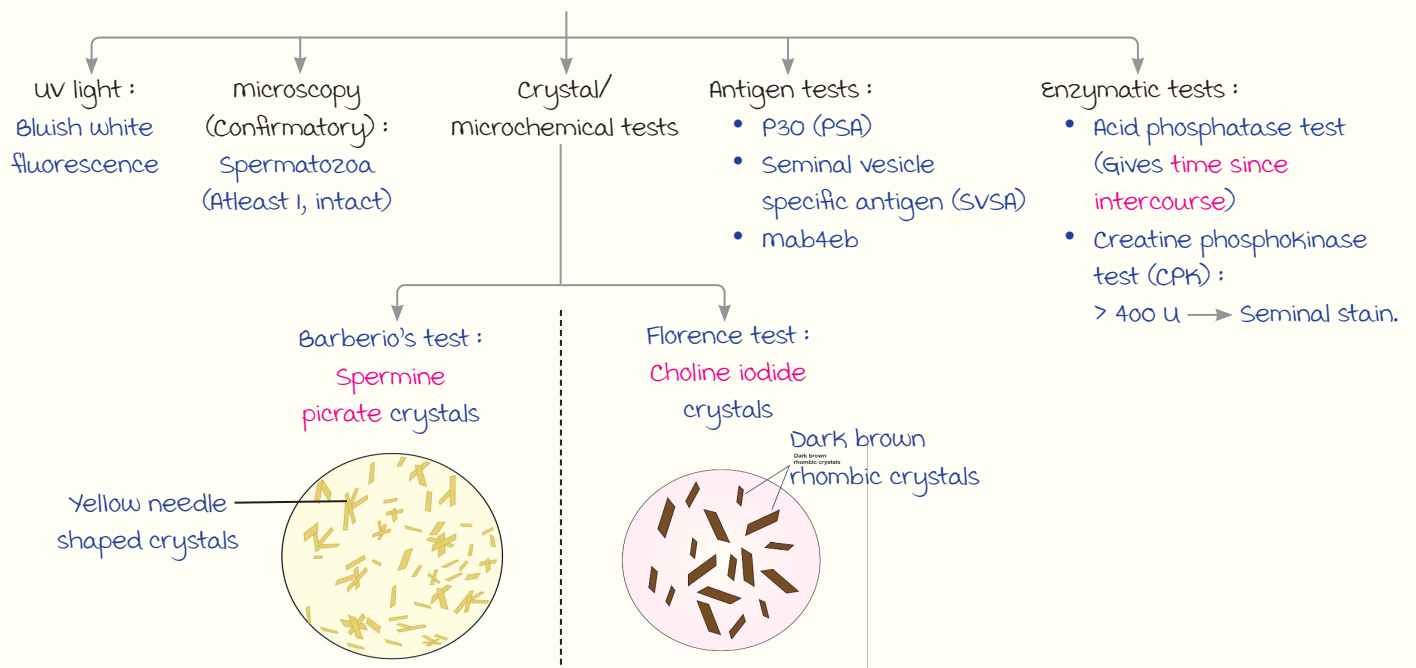
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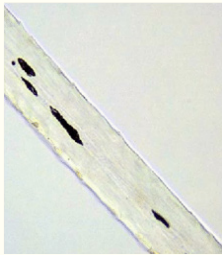
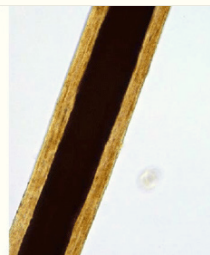


Teichmann test



Takayama test

Species identification : **Precipitin test.****Tests for Seminal Stains** 💀 :**Hair :**

	Human hair	Animal hair
medulla	- Fragmented - Narrow - Absent in few places 	- wider/thicker - Continuous 
medullary index	< 0.3	> 0.5

TOXICOLOGY

Toxicology : Study of poisons.

Toxinology : Study of toxins (Biological substance).

Identification of Poisons

00:00:29

Based on Smell 🦠 :

Autopsy performed by opening cranial cavity first (Helps identify poison by smell).

Smell	Poison
Garlic	Arsenic/phosphorus/aluminium phosphide/organophosphorous compounds(OPC)
Bitter almond	Cyanide
Burnt rope	Cannabis
Acrid pear	- Paraldehyde - Chloral hydrate (Mickey's Finn/knockout drops) → stupefying agent
Fruity	Alcohol
Rotten egg	Hydrogen sulphide
Kerosene	OPC
Hospital	Phenol (Carbolic acid)
Mousy	Conium maculatum (Hemlock) → Causes ascending paralysis
Fishy	Zinc phosphide

Based on Appearance of Stomach Mucosa :

Colour/nature 🦠	Poison
Black necrotic	Sulphuric acid (H_2SO_4)/Oil of vitriol
Yellow	Nitric acid
Buffy white leathery	Phenol
Bluish green	- Copper sulphate ($CuSO_4$) → Bluish green vomitus - Sodium amytal
Slate grey	Mercury
Red velvety	Arsenic

Duties of a Doctor in Poisoning Cases

00:05:52

----- Active space -----

Legal Duties :

- Compulsory duty.
- Preserve the evidence.
- Police intimation (33 BNSS).
- Punishments :
 - Disappearance of evidence : [P] 238 BNS.
 - Failure to inform police.
 - False information provided.

Note :

BNS 286 : [P] Negligent conduct with respect to poisons.

Medical Duty :

Done 1st and then legal duties are taken up.

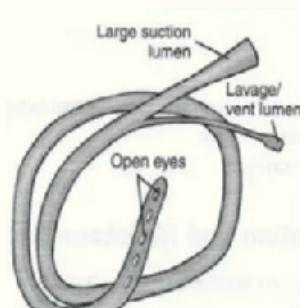
1. Stabilization.

2. Removal of unabsorbed poisons/decontamination 🦠 :

- Gastric lavage : For oral poisons.
- Done using :

Note : 🦠

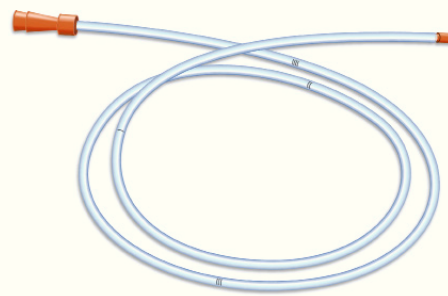
most rapid route of absorption of poison → Inhalation.



Lavacuator



Ewald's tube



Ryle's tube

- Ideal position : Left lateral position.
- Ideal time : < 1 hr.
- Activated charcoal added → Adsorption of poison.
- Absolute contraindication :
 - Corrosives → D/t perforation (except carbolic acid : Leathery mucosa).
- Relative contraindications :
 - Coma.
 - Volatile (kerosene, hydrocarbon) → Aspiration pneumonitis.
 - Convulsant (Strychnos) : Precipitate convulsion.

----- Active space -----

3. Removal of absorbed poisons :

- Dialysis (Hemodialysis > Peritoneal).
- ↑ urinary excretion of poisons :

Alkalization (By IV NaHCO_3) :
For acidic drugs (Salicylates/barbiturates).

Acidification (Rare) : For alkaline drugs.

- multidose activated charcoal.

Antidotes

00:12:15

Poison	Antidote
Arsenic	BAL (Dimercaprol), DMSA
Copper	D-penicillamine, DMSA
Iron	Desferrioxamine
Lead	mild-moderate EDTA
	Severe (Encephalopathy $\oplus$) EDTA + BAL/DMSA
mercury	DMSA, BAL
Cadmium	DMSA
Cocaine	Amyl nitrite
Beta blocker	Glucagon
Carbon monoxide	moderate High flow oxygen
	Severe Hyperbaric oxygen (Causes barotrauma)
Cyanide	Hydroxocobalamin (DOC), Nitrites (Induces methHb)
Digitalis	Digibind
morphine	Naloxone sodium
methanol	Fomepizole + Ethanol
	⊖ Alcohol dehydrogenase
Benzodiazepine	Flumazenil
Acetaminophen	N-acetyl cysteine
OPC	Atropine + oxime
Carbamates	Atropine
Organochlorine (Endrin, DDT)	Only supportive
Pyrethroids	
Oxalic acid ($\downarrow \text{Ca}^{2+}$)	IV calcium gluconate
Hydrofluoric acid ($\downarrow \text{Ca}^{2+}$, $\downarrow \text{Mg}^{2+}$, $\uparrow \text{K}^{+}$)	
Ethylene glycol ($\downarrow \text{Ca}^{2+}$)	

C/i of BAL  :

mnemonic IOC :

- Iron.
- Organic mercury.
- Cadmium.

Eli Lilly's Kit  : Cyanide poisoning

- Amyl nitrite.
- Sodium nitrite.
- Sodium thiosulphate.

morphine poisoning features :

- Coma.
- Pinpoint pupil.
- Respiratory depression.

OP poisoning :

- Oximes can only be given with atropine (worsen poisoning if given alone).
- Atropinisation (Airway secretions are dried & clear) : Stop atropine.

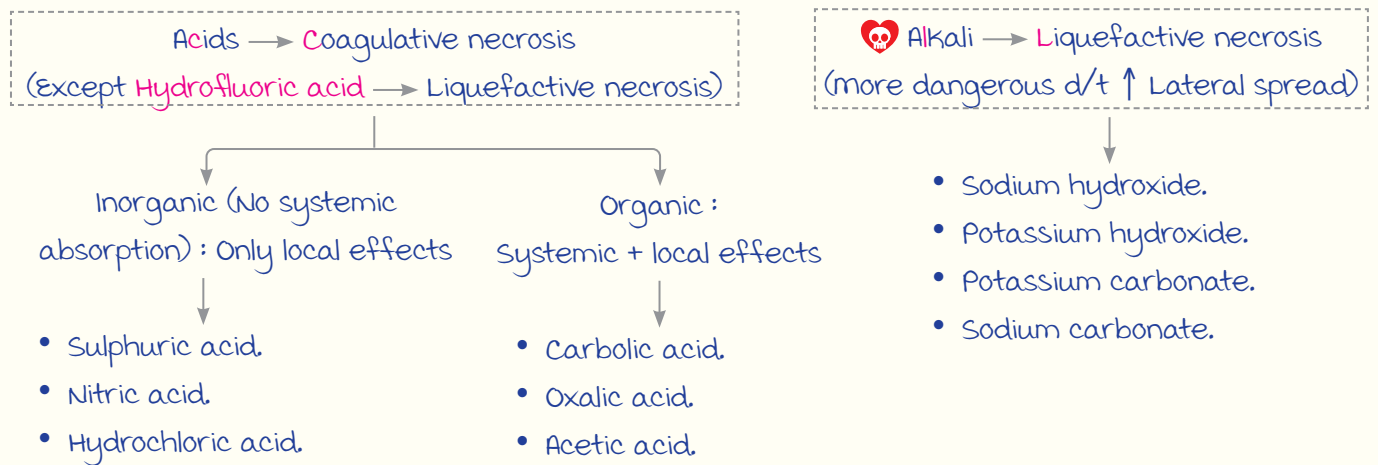
Barium carbonate (Rat poison) :

- White, odourless.
- $\downarrow \text{K}^{+}$.

Corrosives

00:19:00

----- Active space -----



Sulphuric Acid :

Carbonization : Black necrotic tongue, stomach mucosa, black charred skin.

Hygroscopic action : Chalky white teeth.

Nitric Acid ☠ :

Nitric acid + tissue → Picric acid (Xanthoproteic reaction)

Yellow discoloration of skin, teeth, mucosa.

Carbolic Acid/Phenol ☠ :

- Absorbed by all routes.
- Disinfectant : Delay in putrefaction

Ingestion $\xrightarrow[\text{stomach mucosa}]{\text{Leathery/buff white}}$ Liver metabolises

CNS depression,
constricted pupil.

• Pyrocatechol
• Hydroquinone

Urine : Carboluria
(Olive green)

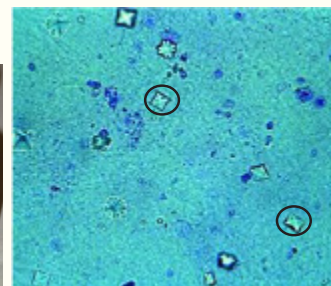
Black tissue deposits : Ochronosis

Oxalic Acid :

- Acid of sugar.
 - Seen in ink/stain remover.
 - Oxalic acid + calcium
(In the blood)
- Hypocalcemia → Tetany.
- Calcium oxalate crystals
↓
Oxaluria (Casts in urine).



Oxalic acid



Dumbbell/Postal-envelope shaped casts

----- Active space -----

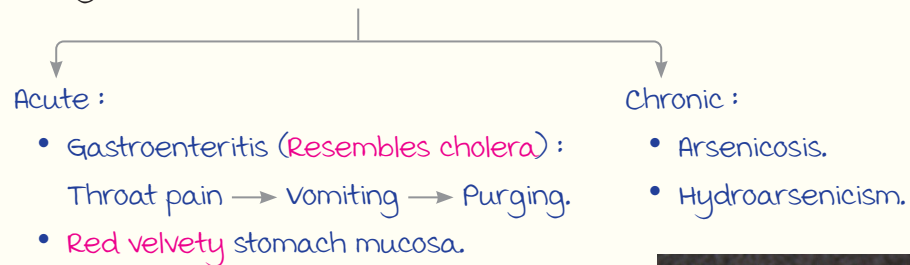
Metallic Irritants

00:23:45

- m/c acute heavy metal poisoning : Arsenic.
- m/c chronic heavy metal poisoning : Lead.
- Lead poisoning route 💀 :
 - Children : Oral.
 - Adult : Occupational inhalation.

Arsenic :

- metalloid.
- most toxic :
 - Form : Arsine (Gas).
 - Salt : Arsenic trioxide (Colourless, odourless, tasteless, water soluble).

Types of poisoning :**Arsenicosis 💀 :**

mnemonic : ARSENIC poisoning.

- Aldrich mees lines : Transverse white line in nail (Also in thallium).
- Alopecia.
- Raindrop pigmentation.
- Skin eruption : Hyperkeratosis in palms & soles.
- Neuropathy :
 - Symmetrical
 - Peripheral
 - Sensory motor polyneuropathy
 - Resembles Guillain-Barre syndrome
 - Also in Thallium
- Ischemia (Periphery) : Gangrene (Black foot disease).
- Carcinogenic.
- Pancytopenia : Bone marrow suppression.



Aldrich mees line



Raindrop pigmentation

Samples 💀:

- Acute poisoning : **Urine** (Reliable) > Blood.
- Chronic poisoning : Bone, nail, hair.
- Detected in exhumed bodies, cremated remains, decomposed bodies.

Tests :

- march test.
 - Reinsch test.
 - Neutron activation analysis (NAA)
 - Atomic absorption spectrometry (AAS)
- } New.

----- Active space -----

Note :

Poison detected in decomposed bodies even without consumption → Alcohol.

mercury :

Occupation : Hatters, glassblowers.

Types :

- Elemental :
 - Not orally absorbed.
 - Vapour : Toxic.
- Compound :
 - Toxicity : Organic (methyl Hg) > Inorganic (mercuric > mercurous salts)

Chronic toxicity/hydrargyria 💀:

Mnemonic : **METAL**

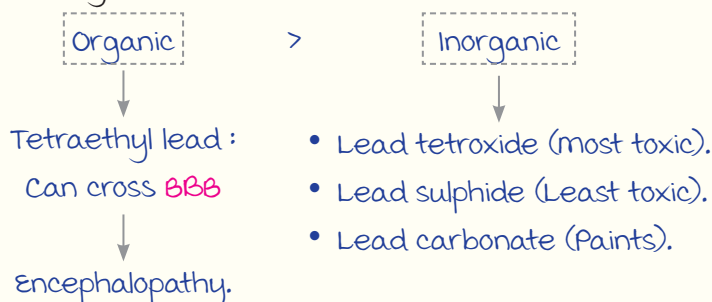
- **minamata disease** : Fish consumption (**methyl Hg**).
- **Erethism** : Neuropsychiatric (**mad hatter syndrome**).
- **Tremors** : Coarse intentional tremors (**glassblowers/hatters shakes**).
- **Acrodynia** : Pink disease (**Calomel/swift fever disease**).
- **Lentis** : Anterior lens capsule deposit.
- Pink, peeling.
- Painful puffy peripheries.
- Pruritic.
- usually paediatric.



Acrodynia

Lead 💀:

Toxicity :



----- Active space -----

Chronic lead poisoning/Saturnism/Plumbism :

mnemonic : ABCDEFG

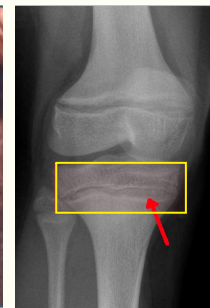
- Anemia : ⊖ Heme synthesis → microcytic hypochromic anemia.
(⊖ ALA dehydratase, ⊖ Coproporphyrinogen oxidase, ⊖ Ferrochelatase).
- Basophilic stippling of RBCs : ⊖ 5-pyrimidine nucleotidase.
- Burtonian lines : Bluish deposition in gums of caries teeth.
- Bone line (Calcium deposition) : metaphyseal dense opacity.
- Colics, constipation : Painter's colic/Saturnine colic/dry belly ache.
- Cabot ring : Inclusion body.
- Drop : Wrist drop, foot drop.
- Encephalopathy.
- Facial pallor : most consistent finding.
- Gout : Saturnine gout.



Basophilic inclusions



Burtonian lines



Bone line

• Investigations :

Blood :

- Lead level.
- Free erythrocyte protoporphyrin (FEP).
- Zinc protoporphyrin.

>

Urine :

- Lead level.
- ALA level ↑↑ (Screening for workers).
- Urine porphyrin level.

Note :

Bluish deposition in gums also seen in mnemonic : Blue Slim Card

- Bismuth.
- Silver.
- Iron.
- Mercury.
- Copper.

Cadmium :

----- Active space -----

Itai itai/Ouch ouch disease :

- Yellow ring in teeth.
- Cadmium replaces Ca^{2+} from bones → Bone deformities.
→ Pathological fractures → ↑↑ Bone pain.

Thallium 🦠 :

Poisoning :

mnemonic : All Nerves Burn

- Alopecia with madarosis.
- Aldrich mees line.
- Sensory motor poly Neuropathy.
- Behavioural changes.

Seen only in thallium poisoning (Not in arsenic)

Treatment : Prussian blue.

Non-Metallic Irritants

00:39:08

White Phosphorous :

- Toxic.
- Luminous.
- Emit fumes : Spontaneous combustion.
- Deep burns on contact (Rx : Topical CuSO_4).



white phosphorous

Acute poisoning :

Stage 1 : Gastroenteritis

- vomitus & stool luminescent
- Smoky stool syndrome

Stage 2 : Asymptomatic

→ Patient discharged & dies later

Stage 3 : Hepatic failure/
multiorgan failure

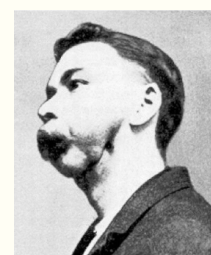
medical negligence.



Red phosphorous : Non-toxic

Chronic poisoning :

- Phossy jaw/Glass jaw/Lucifer jaw.
- Occupational inhalational disease.
- Osteomyelitis → Osteonecrosis of jaw :
 - Painful.
 - Swelling.
 - multiple discharging sinus : Garlicky odor.



Phossy jaw

----- Active space -----

Aluminium Phosphide :

- Rat poison (Celphos).
- Available as tablets Release → **Phosphine gas** (On contact with HCl in stomach).

Investigations :

- Silver nitrate breath test
 - Silver nitrate test
- } → Filter paper turns **black**.

Treatment :

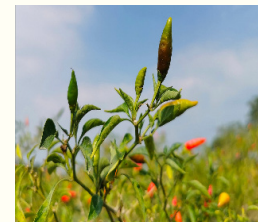
- No specific antidote.
- Supportive management.
- mgSO_4 (For **↓mg**).

Plant Toxins

00:41:21

Toxic Seeds 🦠 :

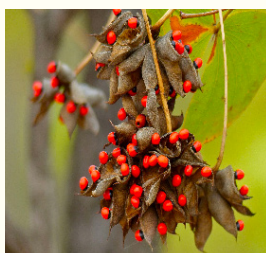
Plant/seed	Active principle	Features
Ricinus communis : Castor/Arandi	Ricin (⊖ protein synthesis)	Crushed seeds : • Oil not toxic • Residue (Toxic) → Bloody diarrhea
Abrus precatorius : • Gunchi • Rati seeds • Rosary • Crab's eye	• Abrin • Abralin • Abrine • Abric acid	• Powdered seeds → Sui needles • Ideal cattle poison (Resembles viperine bite)
Semecarpus anacardium : • Bhilawa • marking nut • Dhobi's nut	• Semecarpol • Bhilawanol	• Crushed seed : Black juice • used for malingering : Artificial bruise (Develop blisters)
Croton tiglium : Jamal gota	• Crotin • Crotonoside • Crotonic acid	Crushed seed : Oil & residue both toxic
Calotropis : madar/Akdo	• Calotropin • Calotoxin • Calactin	All parts toxic
Capsicum annum	• Capsicin • Capsaicin	Hunan's hand syndrome : Contact dermatitis (Occupational)



Capsicum annum



Ricinus communis



Abrus precatorius



Croton tiglium



Semecarpus anacardium



Calotropis

Somniferous Poisons :

Papaver somniferum (Opium) ☠ :

- AKA **a**fm/madak .
- milky exudate
↓ dry
Crude opium
↓
Opioids (morphine, codeine).



Papaver somniferum



Capsule

----- Active space -----

Heroin :

- AKA smack/junk/brown sugar/tar.
- Constituent : **Diacetyl** morphine.
- Semisynthetic derivative of morphine.
- Highly addictive.
- Routes of abuse :
 - Inhaling vapour : Chasing the dragon.
 - Inject into vein : main lining.
 - Inject to skin : Skin popping.



Seeds (**Non-toxic**)

Deliriants :

Mnemonic : **CCD**.

- **C**ocaine.
- **C**annabis.
- **D**atura.

Datura ☠ :

- **muttering delirium**.
- Active principle :
 - **Hyoscine**.
 - Hyoscyamine
 - Atropine.
- **Anticholinergic** symptoms :
 - Dilated pupil.
 - Dried secretions.
 - Delirium.
- Roots & seeds **most toxic**.
- **Stupefying agent** :
 - Railway poison.
 - Roadside poison.
- Antidote : **Physostigmine**.



----- Active space -----



Datura fruit
(Thorn apple) :
Single, large.

vs.



Castor fruit :
Small, cluster.



Datura seeds :

- Dark brown.
- Thick, rough.
- Kidney-shaped.
- Odourless.

vs.



Chilli seeds :

- Yellow.
- Thin, smooth.
- Oval-shaped.
- Pungent smell.

Cannabis :

- AKA Grass/rope/hash/joint/weed/marijuana.
- Active principle : Tetrahydrocannabinol (THC).

Part of plant	Preparation	Active principal %
Dried leaves	Bhang	< 15 %
Dried flower (Female plant)	Ganja (Joint/reefer)	15 - 25 %
Dried resin	Charas/hashish	25 - 40 %
Resin	Hash oil	60 - 70 %



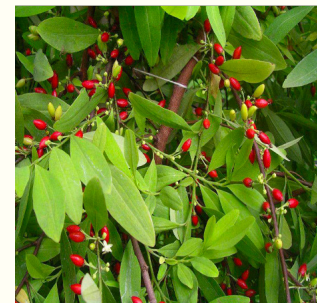
Cannabis

- Chronic cannabis abuse :
mnemonic : AIR.
 - Amotivational syndrome.
 - Insanity : Hashish/hemp insanity
 - Run amok : Homicidal impulse
 - Koro phenomenon/genitalia retraction syndrome (Delusion).
 - Flashback phenomenon : Effects without drug intake (Also seen with LSD).

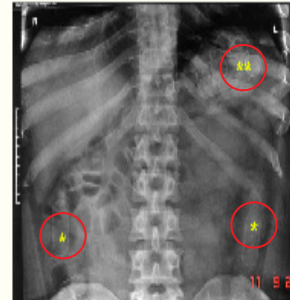
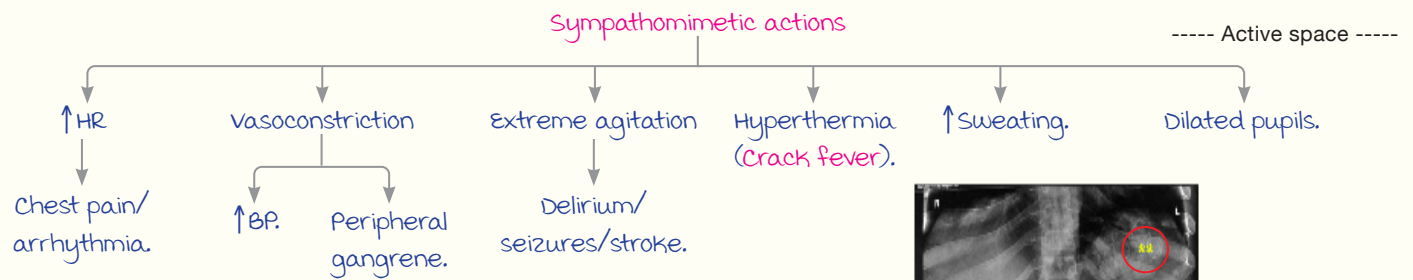
} Not criminally liable : 2a BNS.

Cocaine 🦴 :

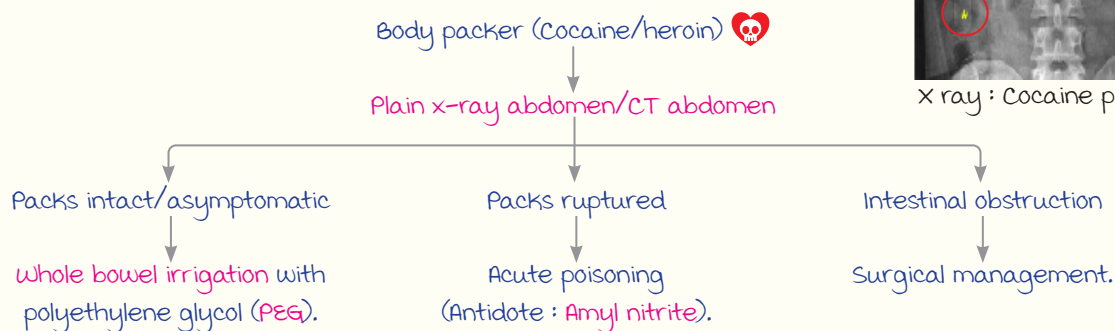
- AKA coke/snow/white lady/she/crack.
- m/c route of abuse : Snorting.
- Speedball :
Cocaine (CNS stimulant) + Heroin (CNS depressant).
- Acute poisoning : Body packer syndrome .



Erythroxylum coca leaves



X ray : Cocaine packets



• Chronic cocaine abuse :

- AKA Cocainism/cocainomania/cocainophagia.

- Triad :

Black staining (Tongue/teeth)

Septal perforation (If snorting) Peripheral gangrene

- Tactile hallucinations : Cocaine bugs/magnan syndrome/Formication.

Note :

123 BNS : Causing hurt using poison with the intent to commit crime.

Inebriants

00:58:07

Order of toxicity : Isopropyl alcohol > methanol > Ethanol.

Ethanol :

- CNS depressant.
- Alcoholic gaze nystagmus : 80 - 100 mg% blood alcohol concentration (BAC).

Stages of alcohol effect :

1. Stage of excitement.
2. Stage of motor incoordination (↑ Incidence of RTA).
3. Stage of coma.

----- Active space -----

Commercial preparations & alcohol content :

- Beer : 6 - 10%.
- Wine.
- Whiskey/brandy : 42 - 45%.
- Rum : 50%.
- Vodka : 50 - 60%.

Absorption :

- 20% → Stomach.
- 80% → Small intestine.

↑ Absorption :

- On an empty stomach.
- With warm water.
- With carbonated drinks.

↓ Absorption :

- High protein meal.
- Cold water/ice.
- Full stomach.

Distribution :

- Effect of alcohol : Females (Smaller water compartment) > males.
- Levels in different compartments :
 - Blood : urine = 1 : 1.3.
 - Blood : CSF = 1 : 1.1.
 - Blood : vitreous = 1 : 1.2.
 - **Blood : Alveolar air** = 1 : 2100 (used in breath analyzers).

Issue of drunkenness certificate :

Smell of alcohol	motor coordination	Opinion
⊖	Normal	Not consumed alcohol
⊕	Normal	Consumed alcohol but no influence
⊕	Abnormal	Consumed alcohol and under influence

Legal limit : **30 mg% BAC.**

- >30 mg% → Drunk & driving → [P] Section 185 of motor vehicle act.

Alcohol withdrawal syndromes :

Time since stoppage	Syndrome	Symptoms
6 - 12 hours	Abstinence syndrome	Tremors (m/c)
12 - 24 hours	Hallucinosiis	-
24 - 48 hours	Rum fits	-
2 - 5 days	Delirium tremens (if crime committed : Not liable under 22 BNS)	• Clouding of consciousness • Altered sensorium • Hallucinations

Complications :

Thiamine deficiency

----- Active space -----

Wernicke's encephalopathy :

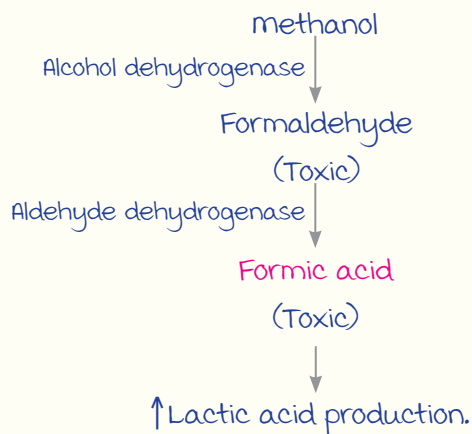
- Global confusion
 - Ophthalmoplegia
 - Ataxia
- } mnemonic : GOA.

Korsakoff's syndrome :

- memory impairment.
- Confabulation.

methanol :

metabolism :



Features 💀 :

- Illicit liquor.
- mass poisoning : Hooch tragedy.
- Symptoms :
 - Blurring of vision
 - Blindness
 } D/t retina & optic nerve involvement.
- High anion gap metabolic acidosis (HAGMA).

Ethylene Glycol :

- Anti-freeze solution.
 - Ethylene glycol → Glycolic acid → Oxalic acid
 - ↳ Hypocalcemia.
 - ↳ urine casts (Oxaluria).
- HAGMA

Spinal And Cardiac Poisons

01:06:02

Spinal Poisons :

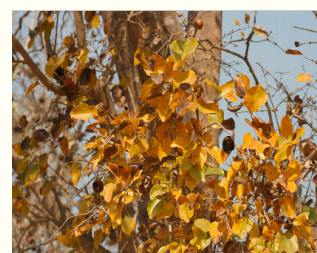
- Strychnos nux vomica.
- Gelsemium.

Strychnos nux vomica :

- AKA Kuchila/Dog's button.
- Even 1 crushed seed is fatal.
- Active principle :
 - Strychnine (most potent).
 - Brucine.
 - Loganin.



Seed



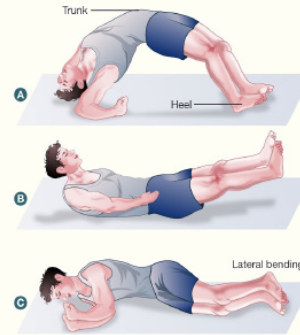
Tree

----- Active space -----

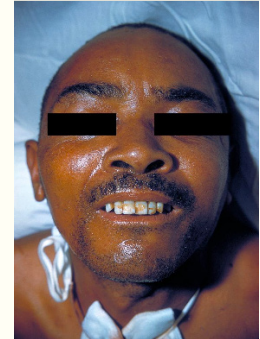
- Acts on **anterior** horn cell → muscle convulsions (Resembles **tetanus**)
(⊖ **Glycine**)
 - Rigor mortis early onset
 - Post-mortem calorificity
 - Delay in putrefaction
 (Autopsy). 🦠

• Presentation :

- Risus sardonius.
- **Opisthotonus** (m/c) : Backward hyperextended spine (A).
- Emprosthotonus : Forward hyperflexed spine (B).
- Pleurosthotonus : Lateral flexion of spine (C).



Abnormal posture



Risus sardonius

Cardiac Poisons :

mnemonic : Car **DONA** (Digitalis, Oleander, Nicotine, Aconite).

Aconite 🦠 :

- AKA blue rocket/**mita zaher**/ Devil's helmet/monk's hood.
- Active principle : **Aconitine**.
- All parts toxic, **most toxic** : **Root**.
- **Perioral paraesthesia**.
- Cardiac arrhythmias.
- **Hippus** (Alternate constriction & dilatation of pupil).



Aconite plant



Roots of aconite

Nerium odorum/Pink oleander :

Active principles :

- Neriodorin.
- Karabin.
- Folinerin.
- Roeganin.



Cerbera thevetia/Yellow oleander :

Active principles :

- Cerberin.
- Nerifolin.
- Thevetin.
- Thevetoxin.
- Peruvoside.
- Ruvoside.



Cerbera odollam :

- Confused with mango → Accidental poisoning.
- Active principles :
 - Cerberin.
 - Cerberoside.
 - Odollin.
 - Odollotoxin.



----- Active space -----

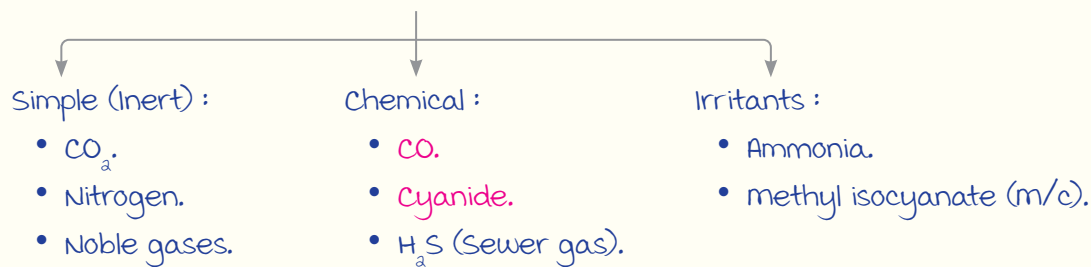
Digitalis/Foxglove :

Rx : Digibind.



Asphyxiants

01:11:17



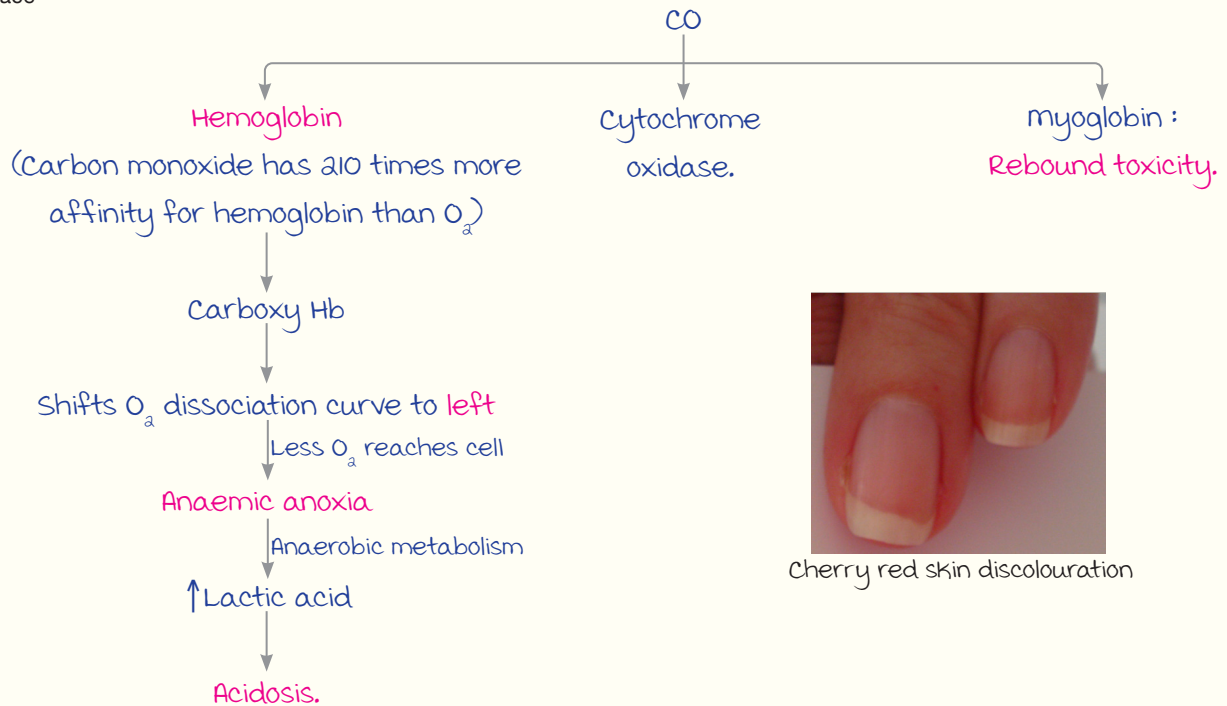
Carbon monoxide 🚫 :

- Odourless.
- Colourless.
- Lighter than air.

Sources :

- Gas heaters (Accidental poisoning).
- Fire inside closed space.
- Garage death (Suicide by running vehicle in closed garage).
- Coal grills at home in a closed space.

----- Active space ----- mechanism of action :



Clinical Features :

Carboxy Hb levels	Symptoms
0 - 10%	No symptoms
10 - 20%	mild headache
20 - 30%	• Throbbing headache • Emotional instability
40 - 50%	• ↑ Confusion • Ataxia • motor incoordination • Cherry red colour skin } mimics drunkenness
50 - 70%	Coma
> 80%	Rapid death from respiratory arrest

Pm findings :

- Cherry red hypostasis.
- Cutaneous blisters.
- B/L symmetrical necrosis of lenticular nucleus.

Treatment :

- High flow O₂.
- Hyperbaric O₂.

Note :

Cutaneous blisters in autopsy seen in :

- CO poisoning.
- Barbiturate poisoning.

Cyanide & H₂S :

----- Active space -----

	Cyanide	H ₂ S
Odour	Bitter almond	Rotten egg
Pathology	⊖ Cytochrome oxidase (⊖ ETC) → Histotoxic anoxia ↓ Lactic acid Lactic acidosis	
Treatment	- Hydroxocobalamin - Eli Lilly's kit	- High flow O_a - Nitrites (Induces methHb)
Colour of hypostasis	Brick (Bright) red	Bluish green

Adulterated Toxins

01:19:12

Argemone mexicana 🍀 :

- Yellow poppy/mexican poppy.
- Mustard oil adulterated with Argemone oil.

Active principles :

- Sanguinarine
 - Dihydrosanguinarine
- } Causes Epidemic dropsy.



Argemone mexicana



Mustard seeds



Argemone seeds

Snake Venom

01:19:56

Ophiology : Study of snakes.

Ophitoxemia : Envenomation (By snake venom).

Venomous vs Non-venomous Snake :

	Venomous	Non-venomous
Head scales	usually small (Except cobra & krait)	usually large
Belly scales	Large & cover entire breadth of belly	Small & never cover entire breadth of belly
Tail	Compressed	Not markedly compressed
Bite	2 fang marks	Small teeth marks in a row
Lives	Nocturnal	Diurnal

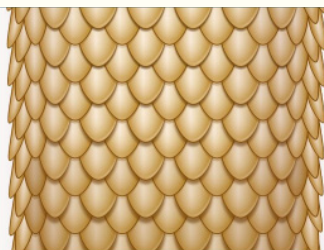
----- Active space -----

Large scaled venomous snake :

- Cobra : 3rd supra labial scale largest.
- Krait : 4th infra labial scale largest.



Venomous snake belly scale



Non-venomous snake belly scale

Venomous Snakes 🩸 :

	Elapidae	Viperidae	Hydrophidae
Snakes	• King cobra (Fatal dose : 12 mg) • Common cobra • Krait (Fatal dose : 6 mg)	• Russell's viper • Saw scaled viper	Sea snake
Venom	Neurotoxic	Haemotoxic	myotoxic
Clinical	Neuromuscular paralysis	Disseminated intravascular coagulation	• myonecrosis/Renal failure • Hyperkalemia (Tall T wave)

Naja naja :
Common cobra.



Ophiophagus hannah :
King cobra (Longest).



Daboia russeli :

- Russell's viper.
- Diamond shape marks, One on top, 2 on the sides.



Echis carinatus :
Saw-scaled viper.



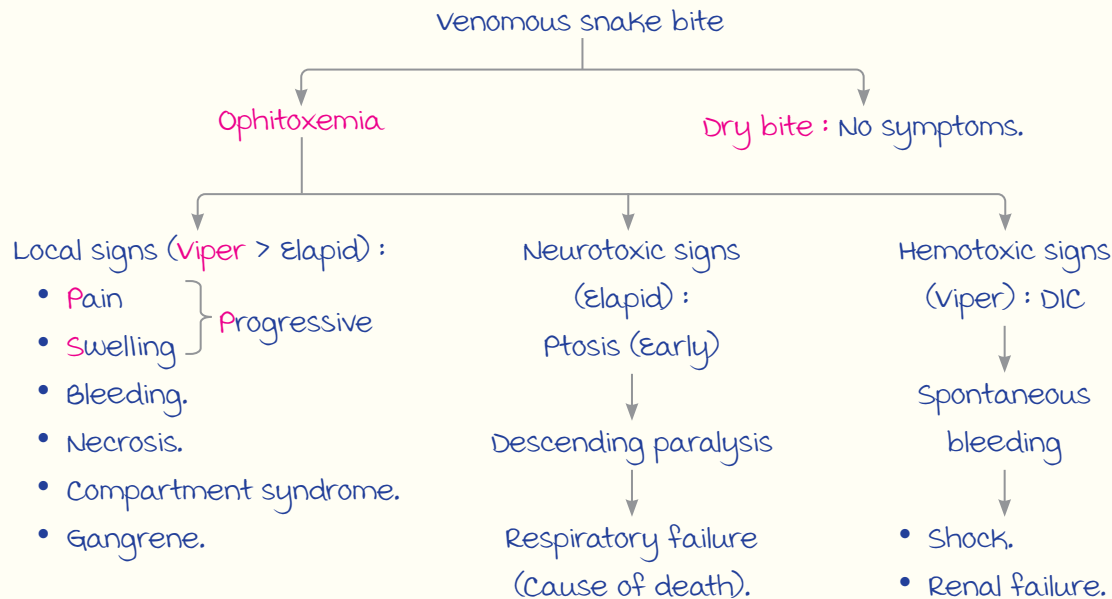
Bungarus caeruleus (Bungarotoxin) :
Common Krait.



Clinical Presentation 🦋:

----- Active space -----

Fear : m/c.



management :

Tests :

- Single breath count test (Bedside test; done in elapid's bite) :
Progressive decline in numbers count per breath → Impending respiratory failure.
- 20 min whole blood clotting test (Viperidae) :
Collect 2 mL blood in a test tube :
 - Normal : Clot within 20 mins.
 - Viper bite : No clot even after 20 mins.

management of Snake Bite :

1st aid :

mnemonic : RIGHT 🦋

- Reassure patient.
- Immobilize entire limb : Crepe bandage/cloth.
 - To occlude lymphatics (55 - 70 mmHg).
 - Never use ligation/incision/tourniquet.
- Get to hospital immediately.
- Tell doctor about any systemic symptoms that manifested on the way.



Ptosis in elapidae bite



Spontaneous bleeding in viper bite

----- Active space -----

At hospital ❤️ :

- Observation
- Anti-snake venom :
 - Only if ophitoxemia.
 - IV 8 - 10 vials.
 - Effective for "Big 4" (Common cobra, krait, Russell's viper, saw-scaled viper).
- IV Neostigmine + atropine → Reversal of neuromuscular paralysis (Cobra bite).

Scorpion & Spanish Fly

01:32:30

Scorpion :

- Indian red scorpion & black scorpion.
- Venom : Similar to snake venom.
↓ Chance of death (D/t ↓ poison injected).
- Bite : Only 1 punctured mark.



Red scorpion

Features ❤️ :

- Pain : m/c clinical feature.
- Autonomic storm :
 - Cholinergic :
 - Sweating, salivation.
 - Priapism.
 - Hypotension.
 - Cold extremities.
 - Adrenergic :
 - Hypertension.
 - Tachycardia → myocardial dysfunction.
 - Pulmonary edema.



Black scorpion

Treatment :

- Prazosin.
- Anti-scorpion venom.

Spanish Fly (Cantharides) :

Powdered form → Aphrodisiac.

Active principle : Cantharidin.

Features :

- Skin : Locally irritant.
 - Redness.
 - Burning sensation.
- Nephrotoxicity.
- Blisters.



Spanish fly

• On oral consumption :

- Haematemesis.
- Burning sensation & pain throughout GIT.
- Dull pain in loin.
- Hematuria.
- Priapism in males.

----- Active space -----

Organophosphorous (OP) Poisoning

01:35:10

Irreversible AchE ⊖.

Cholinergic syndrome (< 1 day) : Acute poisoning.

- | | |
|--|--|
| <ul style="list-style-type: none"> • muscarinic : (Rx : Atropine) mnemonic : DUMBELS <ul style="list-style-type: none"> - Diarrhea. - Urination. - miosis. - Bradycardia. - Bronchorrhea. - Emesis. - Lacrimation (Red tears). - Salivation. | <ul style="list-style-type: none"> • Nicotinic (Rx : Oximes) : - Irritability. - Confusion. - Convulsion. - muscle paralysis. |
|--|--|

Intermediate syndrome (1 - 4 days) :

- **P**roximal muscle weakness
- Cranial nerve palsy

Delayed syndrome (1-4 weeks) :

Distal neuropathies.

Rx : Supportive measures.

Investigations :

- **RBC cholinesterase** (Specific).
- Pseudocholinesterase (Sensitive & easy to perform).

FORENSIC PSYCHIATRY AND LEGAL SECTIONS

Application of psychiatry in administration of justice.

Symptoms of Psychiatric Disorders

00:00:16

Delusion :

Types	Characteristics
Grandeur	Imagination of being rich & powerful
Persecution/Paranoid	Belief that someone is about to cause harm
Reference	Belief that everyone is referring to them
Influence/Control	Belief that one's actions and thoughts are influenced/controlled by some external power
Infidelity/ Othello syndrome	Belief that spouse is not loyal
Nihilistic	- Does not believe in one's own existence or world - Commonly seen in : Depression
Hypochondriacal	- Belief of having illness - medically fit - Frequent visits to hospital
erotomania/ De Clerambault syndrome	- Females > males - Person thinks a person of superior status is in love with them
Capgras syndrome	Belief that a double has replaced a close member
Fregoli's syndrome	Belief that a familiar person disguises as different people

Other Symptoms :

Symptom	Features	Types
Hallucination	- False sense of perception - No external stimuli	- Visual - Auditory - Gustatory - Tactile : magnan's syndrome (Cocaine abuse)
Illusion	False interpretation of stimuli	-
Impulse	- Sudden and irresistible force - Compels a person to do a conscious act - No motive/forethought	- Kleptomania → Steal things of low value - Pyromania → Set things on fire - Trichotillomania → Pull out one's own hair - Dipsomania → Drinking alcohol - Oniomania → Shopping

Symptom	Features	Types
Phobia	Excessive/irrational fear of a particular object/situation	- Acrophobia → Height - Agoraphobia → Crowd - Claustrophobia → Closed spaces - Nyctophobia → Darkness - Mysophobia → Germs

----- Active space -----

Lucid interval : Period of sanity between episodes of mental illness.

Civil & Criminal Responsibilities of Insane

00:05:20

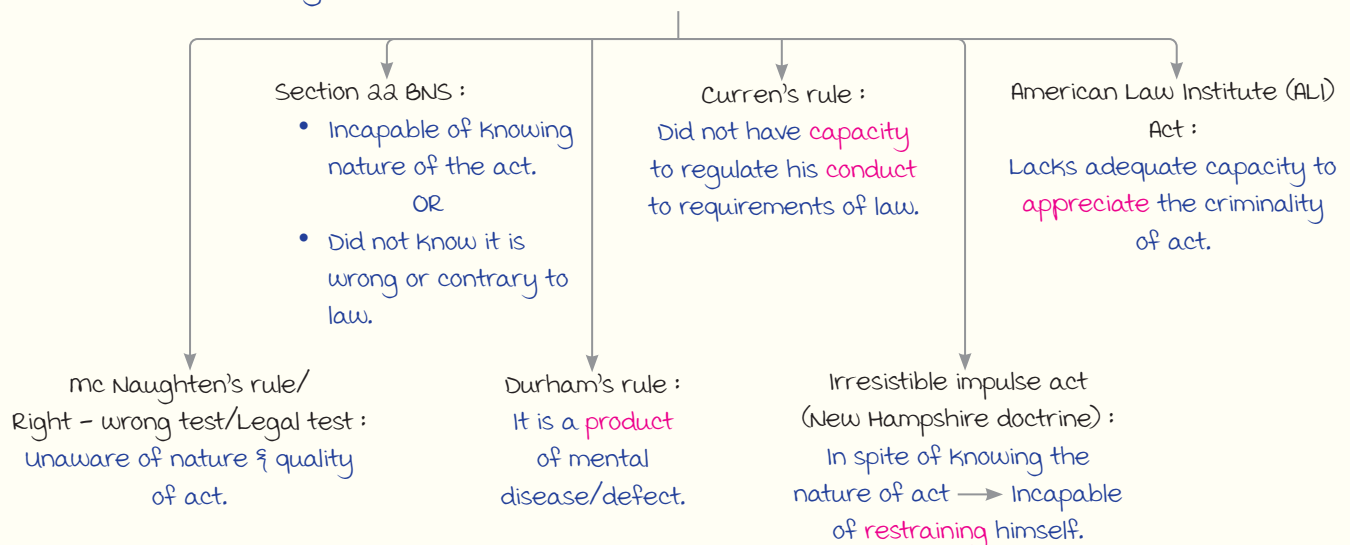
Civil Responsibilities :

Testamentary capacity :

- Capacity of a person to make a valid will.
- Holographic will : Will handwritten by self.

Criminal Responsibilities :

- Law presumes everyone is sane.
- Onus of proof of mental illness 🦴 : On accused.
- Complete psychiatric evaluation to be done : 367 BNSS.
- Procedure to deal with accused persons of unsound mind : 368 and 369 BNSS.
- An accused person (with unsound mind) is not criminally responsible if at the time of committing an act :



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BNS (Bharatiya Nyaya Sanhita)

00:08:27

	BNS	Explanations
Injury	Section 2 (14) [D]	Any harm done to : • Body • mind • Property • Reputation
Criminal responsibility	20	Crime done by person <7 years → Not liable
	21	Crime done by person 7-12 years → Punishable if mentally mature
	22	Crime done by insane person → Not liable
	23	Crime done under involuntary drunkenness → Not liable
	24	Crime done under voluntary drunkenness → Liable
Consent	30	Emergency life saving situations → Consent not mandatory
Rape victim	200 [P]	Non-treatment of victims of rape/vitriolage
	72 [P]	Prohibits disclosure of identity of rape victim
	184 BNSS	medical examination of rape victim
	397 BNSS	Treatment of victim of rape/POCSO/vitriolage
male offences	74 [P]	Indecent assault (Act outraging modesty of a woman)
	75 [D & P]	Sexual harassment of a woman
	76 [D & P]	using force to disrobe a woman
	77 [D & P]	voyeurism
	78 [D & P]	stalking
	80 [P]	Dowry death (within 7 years of marriage)
Homicide	100 [D]	Culpable homicide
	101 [D]	Culpable homicide : • Amounting to murder • Not amounting to murder
	103 [P]	murder
	106 (1) [P]	Patient dies d/t medical negligence
	108 [P]	Abetment of Suicide
Evidence preservation	238 [P]	Intentional disappearance of evidence
Hurt 	114 [D]	Hurt
	115 [P]	voluntarily causing hurt
	116 [D]	Grievous hurt
	117 [P]	voluntarily causing grievous hurt
	118 (1) [P]	voluntarily causing hurt with dangerous weapon/means
	118 (2) [P]	voluntarily causing grievous hurt with dangerous weapon/means
	124 [P]	vitriolage (Acid attack)
medical negligence	271 [P]	Negligent act resulting in spread of fatal infection
	272 [P]	malignant act resulting in spread of fatal infection (Intentional)

Note : 💀

Grievous hurt :

1. Emasculation.
 2. Sight of either eye
 3. Hearing of either ear
 4. Member or joint
- } Permanent privation
5. Permanent impairing of power of any member/joint.
 6. Permanent disfigurement of head/face.
 7. Fracture/dislocation of bone/teeth.
 8. Hurt which :
 - Endangers life
 - Severe bodily pain or unable to follow ordinary pursuits for 15 days.

[P] → Punishment
[D] → Definition

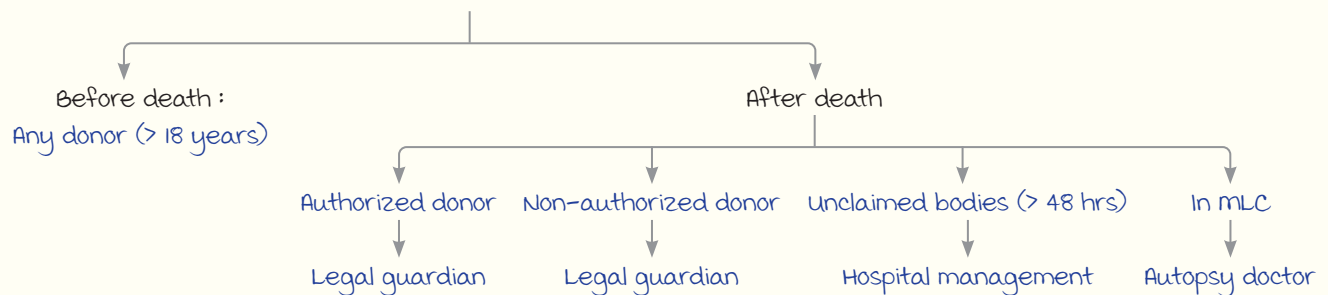
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Transplantation of Human Organs Act 1994 (2014)

00:16:42

- Regulates : Removal, storage & transplantation of human organs for therapeutic purposes.
- Done at : Registered hospitals (Exception : Eye donation).

Authorization for Organ Harvesting :



Types of Organ Harvesting :

- Live donation.
- Cadaveric donation (After brainstem death).

Brainstem Death :

Declaration by medical board :

- Treating doctor
- Hospital in-charge (RMP)
- Independent RMP specialist
- Neurologist/neurosurgeon (If not available : Anaesthetist/intensivist)

Test within
6 hrs

Criteria :

- Rule out reversible causes (Hypothermia/electrolyte imbalance/poisoning)
- Confirm irreversible coma
- Absence of cranial reflexes
- Apnea test

- Transplant surgeon → Never a part of the team.

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Violation :

Doctor involved → Penal erasure (By state medical council).

Protection of Children from Sexual Offences (POCSO) Act 2012 00:19:58

Child <18 years (Irrespective of gender).

Types of Offences :

Penetrative sexual assault (PSA)	Sexual assault (SA)	Sexual harassment	Pornography
Penetration of penis, any object or applying mouth ↓ To vagina, mouth, urethra or anus of a child	With sexual intent : • Touching private parts • Physical contact without penetration • Hormones given to children for early sexual maturity	With sexual intent : • Utter any word/sound/ make gestures • Exhibits any object • Exposes body part • Makes child to exhibit body • Showing pornography	Use of child for pornographic purposes

Aggravated act : If PSA/SA done by (mnemonic : **TRAIT**)

- Person of trust (Parent/guardian/police officer/staff in educational institute).
- Repeat offender.
- Victim age < 12 years.
- Insane victim.
- Two or more people (Gang).

Legal Responsibilities :

Police intimation : mandatory (Failure to report → Punishable).

Medical examination : 184 BNSS.

- In presence of parent/guardian/any person of trust.
- If girl child, only female doctor can examine.